

**EVALUATION OF THE GENERALLY RECOGNIZED AS
SAFE (GRAS) STATUS OF
LYCOPENE PRODUCED BY GENETICALLY
ENGINEERED *SACCHAROMYCES CEREVISIAE*
AS A FOOD INGREDIENT**

Wuhan Hesheng Technology Co., Ltd.
East Lake New Technology Development Zone, Wuhan
CHINA

January 2025

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1. PART I- SIGNED STATEMENTS AND CERTIFICATION.....	4
1.1. Basis of Conclusion:	4
1.2. Name and address of organization:	4
1.3. Name of substance:	4
1.4. Intended conditions of use:	4
1.5. Statutory Basis for GRAS conclusion:.....	5
1.6. Exemption from Premarket approval requirements:	5
1.7. Availability of data and information:.....	5
1.8. Data exempt from Disclosure:	5
1.9. Certification:	5
1.10. Name, position/title of responsible person who signs dossier and signature:	6
1.11. FSIS/USDA – Use in Meat and/or Poultry:	6
2. PART II- IDENTITY AND TECHNICAL INFORMATION.....	7
2.1. Description and Characterization.....	7
2.2. Specifications and Identity	7
2.3. Manufacturing Process.....	9
2.4. Identity of Production Strain.....	12
2.5 Residual Analysis of Protein, DNA, and the Manufacturing Strain	14
3. PART III- DIETARY EXPOSURE.....	16
3.1. Intended Uses and Food Categories.....	16
3.2. Estimated Daily Intake from the Proposed Uses	17
3.3. Natural Sources and Intake	18
4. PART IV- SELF LIMITING LEVELS OF USE	20
5. PART V- EXPERIENCE BASED ON COMMON USE IN FOODS BEFORE 1958	21
6. PART VI- NARRATIVE.....	22
6.1. Data Pertaining to Safety	22
6.2.1. Absorption, Distribution, Metabolism and Excretion of Lycopene	23
6.2.2. Specific Corroborative Studies	24
6.2.2.1. Acute Toxicity of Lycopene Preparation	24
6.2.2.2. Mutagenicity and Genotoxicity Studies of Lycopene Preparation	25
6.2.2.2.1. Bacterial Reverse Mutation Test of Lycopene Preparation	25
6.2.2.2.2. In Vitro Mammalian Chromosome Aberration Test of Lycopene Preparation	25
6.2.2.2.3. In Vivo Mammalian Micronucleus Test of Lycopene Preparation.....	26
6.2.2.3. Summary of Specific Toxicity Studies of Lycopene Preparation.....	27
6.2.3. Animal Toxicity Studies	27
6.2.4. Mutagenicity and Genotoxicity Studies.....	31
6.2.5. Human Studies with Lycopene	32

6.2.5.1. Adverse Effects in Humans	35
6.2.6. Uses and Regulatory Status	35
6.2.6.1. FDA Review of Lycopene GRAS Notices	36
6.2.7. Safety of Production Strain.....	37
6.3. GRAS Panel Evaluation, Summary and Discussion.....	38
6.4. Expert Panel Conclusion.....	42
7. PART VII- LIST OF SUPPORTING DATA AND INFORMATION.....	43

1. PART I- SIGNED STATEMENTS AND CERTIFICATION

In accordance with 21 CFR § 170 Subpart E consisting of sections § 170.203 through § 170.285, Wuhan Hesheng Technology Co., Ltd. (hereinafter Hesheng Technology) hereby informs the FDA that lycopene produced by genetically engineered *Saccharomyces cerevisiae* as a Food Ingredient, is not subject to the premarket approval requirements of the Federal Food, Drug, and Cosmetic Act based on Hesheng Technology's view that the notified substance is Generally Recognized as Safe (GRAS) under the conditions of its intended use described below.

1.1. Basis of Conclusion:

This GRAS conclusion for the use of lycopene produced by genetically engineered *Saccharomyces cerevisiae* has been reached in accordance with the requirements in 21 CFR 170.220.

1.2. Name and address of organization:

Wuhan Hesheng Technology Co., Ltd.
Room D102, Building B5, Building B4-B8,
Wuhan National Biological Industry (Jiufeng Innovation) Base,
No. 666 Gaoxin Avenue,
East Lake New Technology Development Zone, Wuhan
CHINA

1.3. Name of substance:

The name of the substance of this GRAS assessment is lycopene.

1.4. Intended conditions of use:

Hesheng Technology intends to market lycopene produced by genetically engineered *S. cerevisiae* strain JZL03 (hereafter referred as lycopene preparation) for use as a food ingredient and as a nutrient supplement in conventional foods such as baked goods and baking mixes, beverages and beverage bases, breakfast cereals, cheeses, condiments and relishes, confections and frostings, fats and oils, frozen dairy desserts and mixes, gelatins, puddings, and fillings, gravies and sauces, hard candy, milk products, plant protein products, processed fruits and fruit juices, snack foods, soft candy at levels up to 50 ppm and in soups and soup mixes that do not contain tomatoes at levels up to 575 ppm. The proposed uses and use levels are same as in GRN 173 for lycopene derived from *Blakeslea trispora*. It is recognized that there are Standard of Identity requirements for some of these specified foods and these foods will not be referred by their commonly recognized names.

While the use of the lycopene preparation in foods may also impart a color to food products, the intended use of lycopene preparation would fall outside the definition of "color additive" for the following reasons: the lycopene preparation is solely added for its nutritional characteristics and thus may constitute an "unimportant color" [21 CFR 70.3(g)] insofar as the appearance, value, marketability, or consumer acceptability is concerned and does not relate to any use of the ingredient as a color additive [21 CFR 70.3(f)].

1.5. Statutory Basis for GRAS conclusion:

This GRAS conclusion is based on scientific procedures in accordance with 21 CFR 170.30(a) and 170.30(b).

1.6. Exemption from Premarket approval requirements:

Hesheng Technology has concluded that lycopene preparation is not subject to the premarket approval requirements of the Federal Food, Drug, and Cosmetic Act based on our conclusion that lycopene preparation meeting the specifications cited herein, when used as a food ingredient in conventional food products, is GRAS and is therefore exempt from the premarket approval requirements.

It is also our opinion that other qualified and competent scientists reviewing the same publicly available toxicological and safety information would reach the same conclusion. Therefore, we have also concluded that lycopene preparation, when used as described in this dossier, is GRAS based on scientific procedures.

1.7. Availability of data and information:

The data and information that are the basis for this GRAS conclusion will be made available to FDA in a form in accordance with that requested under 21 CFR 170.225(c)(7)(ii)(A) or 21 CFR 170.225(c)(7)(ii)(B) upon request by contacting

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1.8. Data exempt from Disclosure:

Parts II through VII of this GRAS notification do not contain any data or information that is exempt from disclosure under the Freedom of Information Act. There is no privileged or confidential information such as trade secrets and/or commercial or financial information in this document. Therefore, the information contained in this dossier can be made publicly available.

1.9. Certification:

Hesheng Technology certifies that, to the best of its knowledge, this GRAS conclusion is based on a complete, representative, and balanced dossier that includes all relevant information available and obtainable by Hesheng Technology, including any favorable or unfavorable information, and pertinent to the evaluation of the safety and GRAS status of the use of lycopene

preparation. Hesheng Technology accepts responsibility for the GRAS determination that has been made for lycopene preparation as described in this dossier.

1.10. Name, position/title of responsible person who signs dossier and signature:

Name: Yao Zhi
Title: R&D project leader
Wuhan Hesheng Technology Co., Ltd.
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Signature:  Date : 17 / 02 / 2025

1.11. FSIS/USDA – Use in Meat and/or Poultry:

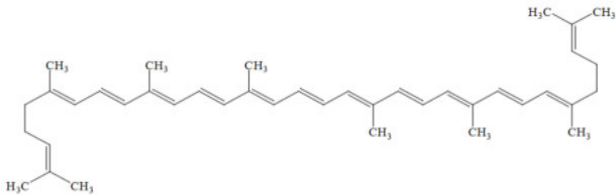
Hesheng Technology does not intend to add lycopene preparation to any meat and/or poultry products that come under USDA jurisdiction. Therefore, 21 CFR 170.270 does not apply.

2. PART II- IDENTITY AND TECHNICAL INFORMATION

2.1. Description and Characterization

The subject of this GRAS assessment is a standardized highly pure ($\geq 96\%$) lycopene preparation, manufactured by genetically engineered *Saccharomyces cerevisiae* strain JZL03. This strain is derived from the *S. cerevisiae* HESHEHG001 strain and has been genetically modified to produce lycopene (see section 2.4). The recombinant strain JZL03 is constructed from the parental strain HESHEHG001. Lycopene is produced by fermentation of genetically engineered *S. cerevisiae* JZL03, followed by extraction and recovery (downstream processing), and final product treatment. It is a red to dark red powder that primarily consists of lycopene ($\geq 96\%$) that is identical to lycopene found in tomatoes. Lycopene is a type of organic pigment called carotenoid. It is related to beta-carotene and gives some vegetables and fruits (e.g., tomatoes) a red color. General descriptive characteristics of lycopene are summarized in Table 1. The sensory requirements shall meet the specifications outlined in Table 1.

Table 1. General Descriptive Characteristics of Lycopene Preparation

Parameter	Description*
Source	Recombinant <i>Saccharomyces cerevisiae</i> strain JZL03
Common name	Lycopene
Natural source	Occur in vegetables and fruits, particularly in tomatoes
CAS number	502-65-8
Chemical structure	
Empirical formula	C ₄₀ H ₅₆
Molecular weight	536.89 Daltons
Melting point	172-175°C
Appearance	Powder
Color	Red to dark red powder
Odor	Characteristic odor
Solubility	Soluble in fats and oils; insoluble in water
Storage	Store under vacuum, away from light, and refrigerated. Recommended to be stored under freezing conditions.

*Based on information provided by Hesheng Technology and other publicly available

2.2. Specifications and Identity

In order to ensure a consistent and safe product, Hesheng Technology has established food grade specifications for lycopene preparation (Table 2). The analytical methods used for the qualitative and quantitative analysis of the individual specification parameters are validated for their intended purpose. The purity of lycopene is at least 96% of the product. Specification limits have been established for total lycopene, all trans-lycopene, heavy metals (arsenic, cadmium, lead, mercury), residual solvents and microbiological contaminants. The total lycopene content and trans-lycopene is measured using a validated GB1886.78-2016 Annex A.3 method. To ensure the purity of the final product, upper limits have been established for the raw materials and processing

aids used in the manufacturing, heavy metals, and microbiological parameters. Batch analysis data from three non-consecutive batches is provided in Table 3.

Lycopene is an acyclic, nonpolar hydrocarbon chain, consisting of a C40 isoprenoid skeleton and 2 open-end rings with relative molecular mass of 536.89 Dalton and CAS Registry number 502-65-8. Similar to other carotenoids, each of the polyene chain double bonds of lycopene could exist in cis or trans conformation, resulting in a large number of geometric isomers. However, the vast majority of carotenoids exist in the all-trans configurations. Lycopene contains thirteen double bonds. Naturally occurring carotenoids are in the all-trans-configuration. Storage, cooking, food processing and exposure to light may result in some isomerization of the all trans form to various cis forms including the 5-cis, 9-cis, 13-cis, and 15-cis forms. All-trans form of lycopene predominates in foods, while the cis isomers, along with trans isomers, are common in human blood and tissue. Specifically, in human plasma all-trans-lycopene and 5-cis-lycopene have been identified as the most abundant lycopene isomers (Hadley et al., 2003). All-trans lycopene is a red crystalline powder with a melting point of 172-175°C that is soluble in fats and certain organic solvents but insoluble in water ethanol and methanol. Lycopene is classified as a non-provitamin A carotenoid, with other examples being lutein and zeaxanthin.

Table 2. Food grade Specifications of Lycopene

Parameters	Limits	Test methods
Appearance	Red to dark red powder	Visual
Total lycopene, ω/%	≥ 96.0	GB1886.78-2016 Annex A.3, HPLC
All-trans-lycopene, ω/%	≥ 90.0	GB1886.78-2016 Annex A.3, HPLC
Carotenoids	Pass the test	A.12, Specification for Lycopene, Announcement No. 6 Issued by Ministry of Health of China in 2012
Loss on drying, ω/%	≤ 0.5	GB 5009.3 Method II, Vacuum Drying
Ethanol/%	≤ 0.5	0861 Determination of Residual Solvents, ChP 2020
Ethyl acetate/%	≤ 0.5	0861 Determination of Residual Solvents, ChP 2020
Solubility	Pass the test	A.11, Specification for Lycopene, Announcement No. 6 Issued by Ministry of Health of China in 2012
Determination of maximum absorption peak ($\lambda = 470$ nm)	Pass the test	A.13, Specification for Lycopene, Announcement No. 6 Issued by Ministry of Health of China in 2012
Lead /mg/kg	≤ 1.0	GB 5009.12, ICP-MS
Arsenic /mg/kg	≤ 1.0	GB 5009.11, HG-AFS
Cadmium /mg/kg	≤ 1.0	GB 5009.15, ICP-MS
Mercury /mg/kg	≤ 1.0	GB 5009.17, ICP-MS
Aerobic plate count/(CFU/g)	≤ 1000	GB 4789.2
Mold and yeasts/(CFU/g)	≤ 50	GB 4789.15
Coliforms/(MPN/g)	≤ 0.3	GB 4789.3
<i>Staphylococcus aureus</i> /25 g	Absent	GB 4789.10
<i>Salmonella</i> /25 g	Absent	GB 4789.4

1. ChP = Chinese Pharmacopoeia; CFU = colony forming unit; MPN = most probable number; GB = Chinese National standard (Guojia Biaozhun).
2. Any commercial preparations of lycopene should be made using lycopene that meets this specification as the raw material, with the addition of food grade raw materials and additives permitted under U.S. regulations.

Analytical data from three non-consecutive representative batches of lycopene preparation (as summarized in Table 3) demonstrate compliance with the product's physical, chemical and microbiological specifications and the ability of the method of manufacture to produce a consistent product.

Table 3. Batch Analysis Data of Lycopene

Parameters	Test results		
	202307A	02020240410	02020240506
Appearance	Dark red powder	Dark red powder	Dark red powder
Total lycopene, ω/%	98.6	98.5	99.3
All-trans-lycopene, ω/%	97.5	98.1	98.7
Carotenoids	Pass the test	Pass the test	Pass the test
Loss on drying, ω/%	0.13	0.084	0.10
Ethanol/%	Not detected	Not detected	Not detected
Ethyl acetate/%	0.0124	0.0127	0.0113
Solubility	Pass the test	Pass the test	Pass the test
Determination of maximum absorption peak ($\lambda = 470$ nm)	Pass the test	Pass the test	Pass the test
Lead /mg/kg	0.0952	Not detected	0.0953
Arsenic /mg/kg	Not detected	Not detected	Not detected
Cadmium /mg/kg	Not detected	Not detected	Not detected
Mercury /mg/kg	Not detected	Not detected	Not detected
Aerobic plate count/(CFU/g)	90	160	120
Mold and yeasts/(CFU/g)	<10	<10	<10
Coliforms/(MPN/g)	<0.3	<0.3	<0.3
<i>Staphylococcus aureus</i> /25 g	Absent	Absent	Absent
<i>Salmonella</i> /25 g	Absent	Absent	Absent

Limit of detection (LOD) or Limit of quantitation (LOQ) for 'not detected' tests are as follows: Lead: 0.05mg/kg; Arsenic: 0.010mg/kg; Cadmium: 0.005mg/kg; Mercury: 0.003mg/kg; Ethanol:100mg/kg.

2.3. Manufacturing Process

Lycopene is produced by controlled fed-batch submerged fermentation of *Saccharomyces cerevisiae* strain JZL03, constructed from *S. cerevisiae* HESHEHG001, in accordance with Current Good Manufacturing Practices (cGMP) for Food and the principles of Hazard Analysis and Critical Control Points (HACCP). The production process involves the fermentation process, extraction and purification process, and all measures are taken to comply with cGMP and HACCP.

All raw materials, processing aids, and food contact substances used in the manufacturing process of lycopene preparation are used in accordance with applicable U.S. regulations, are GRAS for their intended uses, or are the subject of an effective food contact notification.

The manufacturing process overview is provided in Figure 1.



Figure 1. Manufacturing Process Overview

2.3.1. Fermentation Process

2.3.1.1. Fermentation process overview



Figure 2. Fermentation Process Overview

2.3.1.1 Detailed Fermentation Process

2.3.1.1.1 Strain plate culture

Withdraw the liquid culture from the strain preservation tube and streak onto YPD plates (glucose, peptone, yeast extract, and agar). Incubate at 30°C until monoclonal colonies develop.

2.3.1.1.2. Primary seed culture

Select single colonies from the plate and inoculate into YPD medium (glucose, peptone, yeast extract) for primary seed culture. Transfer the culture once it becomes turbid (OD_{600} greater than 30).

2.3.1.1.3. Secondary seed culture

Transfer the primary seed culture into secondary YPD medium for secondary culture. Transfer the culture once it becomes turbid (OD_{600} greater than 30).

2.3.1.1.4. Seeding tank culture- 100 L

Transfer the secondary seed culture into a 100 L LGM seeding tank (glucose, ammonium sulfate, potassium dihydrogen phosphate, and magnesium sulfate heptahydrate). Ferment at certain temperature with a controlled temperature; OD_{600} greater than 30.

2.3.1.1.5. Seeding tank culture- 5000 L

Introduce the strain culture from the 100 L seeding tank into a fermentation tank containing 3000 L of substrate (yeast extract, glucose, and magnesium sulfate heptahydrate). Ferment at certain temperature with a controlled temperature. Continuously monitor ethanol/glucose levels

within the tank, and cease fermentation once the yield stabilizes. Subsequently, extract and purify the fermented broth to obtain lycopene.

2.3.2. Extraction and Purification Process

2.3.2.1. Extraction and purification process overview

The process flow for extracting and purifying lycopene from the fermentation broth is as follows:

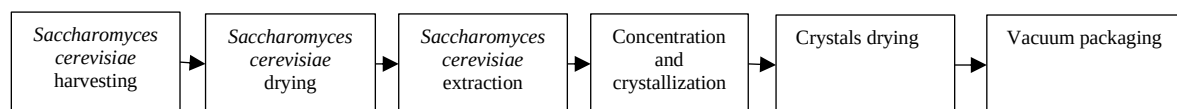


Figure 3. Extraction and Purification Process Overview

2.3.2.2 Detailed Extraction and Purification Process

2.3.2.2.1. *Saccharomyces cerevisiae* harvesting

Centrifuge the fermentation broth to collect wet biomass.

2.3.2.2.2. *Saccharomyces cerevisiae* drying

The wet biomass is dried at high temperature to obtain dry biomass powder.

2.3.2.2.3. *Saccharomyces cerevisiae* extraction

Stir the biomass with 20 times its volume of ethyl acetate at 20°C for half an hour, then filter and collect the filtrate. Stir the filter cake again with 20 times its volume of ethyl acetate at 20°C for another half hour, filter, and collect the filtrate. Combine the filtrates.

2.3.2.2.4. Concentration and crystallization

Concentrate the extract to 4 times its volume, further crystallize overnight at 5°C. Filter and wash the crystals with ethanol.

2.3.2.2.5. Crystals drying

Transfer the crystals to a vacuum drying chamber. Dry at 50°C for several hours under vacuum conditions.

2.3.2.2.6. Vacuum packaging

Once tested to be qualified, place the product into vacuum bags, remove air by vacuum sealing, seal away from light, and store in dry, cool, or refrigerated conditions.

2.4. Identity of Production Strain

2.4.1 Parental (Host) Strain

The production strain used in the manufacturing process of lycopene is *Saccharomyces cerevisiae* HESHEHG001. It's taxonomic lineage is:

Kingdom: Fungi
Phylum: *Ascomycota*
Class: *Saccharomycetes*
Order: *Saccharomycetales*
Family: *Saccharomycetaceae*
Genus: *Saccharomyces*
Specie: *Saccharomyces cerevisiae*
Strain: *Saccharomyces cerevisiae* HESHEHG001

2.4.2 Production strain

For the production of lycopene, the production strain, *S. cerevisiae* JZL03, was constructed from *S. cerevisiae* HESHEHG001 using homologous recombination techniques. A total of 13 genes were integrated into the recombinant *S. cerevisiae* JZL03, including 8 target genes and 5 marker genes. Among the 8 target genes, 3 were exogenous genes, while the other 5 were endogenous genes from *S. cerevisiae*, with their copy numbers increased to enhance expression. These transferred genes were integrated into specific locations on the host strain's genome, replacing existing gene coding regions and non-coding sequences.

All the exogenous genes were *de novo* synthesized *in vitro* before being introduced into the host strain, and were based on well-known annotated genomes from the respective donor organism, without introducing any other characteristics of the donor organism. This approach ensures that the potential toxicity, anti-nutritional properties, allergenicity, and pathogenicity of the donor organisms are irrelevant, and do not pose any associated risks to the final lycopene product.

The target genes play defined roles in the biosynthesis of lycopene, while the marker genes are only used during the construction of the production strains. During the actual production of lycopene, no antibiotics are used in the culture medium of the production strain. Therefore, no antibiotics will be present in the final product. In addition, the final product lycopene does not contain viable manufacturing strain or residual DNA or residual protein fragments from the production strain (as explained in Part 2.5). The Food and Agriculture Organization of the United Nations/World Health Organization (FAO/WHO) Consultation on Biotechnology and Food Safety noted that the presence of antibiotic resistant genes is not a concern if it is not present in foods or released intentionally into the environment (Joint FAO/WHO, 1996).

Table 4 Recombinant *S. cerevisiae* JZL03: Exogenous/Marker Gene Functions and Donor Organisms

Introduced Gene	S.N.	Function	Donor
Exogenous Genes	1	Encodes geranylgeranyl diphosphate synthase	<i>Pantoea ananatis</i>
	2	Encodes phytoene synthetase.	<i>Pseudoscherichia vulneris</i>

	3	Encodes phytoene desaturase.	<i>Blakeslea trispora</i>
Marker Genes	1	Encodes N-(5'-phosphoribosyl) anthranilate isomerase.	Commercial Plasmids pRS424
	2	Encodes 3-isopropylmalate dehydrogenase.	<i>S.cerevisiae</i> S288C
	3	Encodes imidazole glycerol phosphate dehydratase.	Commercial Plasmids pRS423
	4	Encodes aminoglycoside phosphotransferase	Commercial Plasmids pUG6
	5	Encodes orotidine 5'-phosphate decarboxylase	Commercial Plasmids p426 GAL

2.4.3 Biosynthetic Pathway of Lycopene

S. cerevisiae does not naturally produce lycopene, but it can synthesize farnesyl pyrophosphate (FPP), a precursor for lycopene bio-synthesis. The production of lycopene from FPP requires three enzymes, geranylgeranyl pyrophosphate (GGPP) synthase, phytoene synthase, and phytoene desaturase.

The biosynthetic pathway of lycopene is described as follows:

Acetyl-CoA is converted to acetoacetyl-CoA by acetyl-CoA acetyltransferase, and then to HMG-CoA (3-hydroxy-3-methylglutaryl-CoA) by HMG-CoA synthase. HMG-CoA reductase then reduces HMG-CoA to mevalonic acid (MVA). After multiple enzymatic steps, MVA is converted to FPP.

FPP is then converted to GGPP under the catalysis of geranylgeranyl pyrophosphate synthase. Phytoene synthase catalyzes the synthesis of phytoene from GGPP. Finally, phytoene is dehydrogenated to lycopene by phytoene desaturase.

2.4.5 Genetic Stability of the Production Strain

The results of genetic stability tests conducted over five transfer generations on recombinant *S. cerevisiae* JZL03 demonstrate consistent stability across target gene integration, transcription levels, protein expression, growth rate, and lycopene production:

(1) Target Gene Integration Stability:

Genome resequencing of recombinant *S. cerevisiae* JZL03 and comparison with the recipient *S. cerevisiae* HESHEHG001 confirmed that the sequences replaced on the JZL03 genome were completely as expected, with no unintended base modifications. Therefore, the target genes at all replacement sites in *S. cerevisiae* JZL03 are genetically stable across at least five transfer generations.

(2) Stability of Target Gene Transcription Levels:

Using the Actin gene as an internal reference, reverse transcription qPCR was performed to assess the transcriptional stability of target genes across five transfer generations of recombinant *S. cerevisiae* JZL03. The results indicated that the relative expression levels of these target genes

remained within the normal variation range of yeast mRNA from the first to the fifth generation, confirming transcriptional stability across these generations.

(3) Stability of Growth Rate and Lycopene Production:

Under the same cultivation conditions, the growth rates of *S. cerevisiae* JZL03 from the first to the fifth generation were consistent, with no statistically significant differences in dry weight at the same time points ($p>0.05$). Similarly, there were no statistically significant differences in lycopene production between the five transfer generations under the same conditions ($p>0.05$). Thus, the growth rate and lycopene production of recombinant *S. cerevisiae* JZL03 are genetically stable across at least five transfer generations.

(4) Stability of Target Gene Translation Levels:

The intergenerational genetic stability of target gene integration, transcription levels, growth rate, and lycopene production in recombinant *S. cerevisiae* JZL03 collectively indicate that the target genes are expressed stably at the translation level.

2.5 Residual Analysis of Protein, DNA, and the Manufacturing Strain

The residual protein levels in three batches of the final product were determined by Bradford method. The results suggest that the residual protein content in the final product is below the quantitation limit of 17 mg/kg. These findings suggest the absence of residual protein in the final product.

Hesheng Technology studied the presence of any residual transferred DNA in final product. For this, a gel electrophoresis PCR detection method was established to verify the presence of 13 transferred genes from recombinant *S. cerevisiae* strain JZL03, with a limit of detection below 10 ng DNA/g of the final product. The results from three batches showed that all transferred gene DNA fragments from recombinant *S. cerevisiae* strain JZL03 in the final product of lycopene were below the limit of detection. These findings suggest that the final product lycopene, produced by recombinant *S. cerevisiae* strain JZL03 does not contain detectable transgenic DNA fragments.

Presence of manufacturing strain was investigated in the final product lycopene by GB 4789.15-2016, First Method (*China National Food Safety Standard Food Microbiological Examination: Enumeration of Molds and Yeasts: Plate count method*). The results showed that no yeast (*S.cerevisiae*) was detected in any of the three batches of lycopene samples tested. Therefore, it can be concluded that there is no detectable residual recombinant *S.cerevisiae* strain JZL03 in the final product lycopene.

The results of residual protein, residual DNA and residual manufacturing strain is summarized in Table 5.

Table 5. Summary of the Residual Protein, Residual DNA and Residual Manufacturing Strain in Lycopene

Parameters	Limits	Test methods	Manufacturing Lot No.		
			202307A	20270720	20230804
Residual protein	≤ 100 mg/kg	In house (Bradford method)	Less than LOQ (<17 mg/kg)	Less than LOQ (<17 mg/kg)	Less than LOQ (<17 mg/kg)
Residual DNA	Negative	PCR Amplification	Negative	Negative	Negative
Residual manufacturing strain (recombinant <i>S. cerevisiae</i> strain JZL03)	<10 CFU/g	GB 4789.15	<10	<10	<10

3. PART III- DIETARY EXPOSURE

3.1. Intended Uses and Food Categories

Hesheng Technology intends to market lycopene preparation for use as a food ingredient and as a source of the nutrient in conventional foods such as baked goods and baking mixes, beverages and beverage bases, breakfast cereals, cheeses, condiments and relishes, confections and frostings, fats and oils, frozen dairy desserts and mixes, gelatins, puddings, and fillings, gravies and sauces, hard candy, milk products, plant protein products, processed fruits and fruit juices, snack foods, soft candy at levels up to 50 ppm and in soups and soup mixes that do not contain tomatoes at levels up to 575 ppm. The proposed uses and use levels are same as in GRN 173 for lycopene derived from *Blakeslea trispora*. The intended uses of lycopene are substitutional to the existing uses. A summary of the intended use of lycopene food categories and maximum use levels is provided in Table 5.

It is recognized that there are Standard of Identity requirements for some of these specified foods and these foods will not be referred by their commonly recognized names. Foods such as meat and poultry products that come under USDA jurisdiction are excluded from the list of intended food uses of lycopene preparation. Additionally, lycopene preparation is not proposed for uses in infant formula.

Given its color, the use of the lycopene preparation in foods may also impart a color to food products. However, the proposed use of lycopene preparation in selected food categories would fall outside the definition of “color additive” for the following reasons: the lycopene preparation is solely added nutrient supplement and thus may constitute an “unimportant color” [21 CFR 70.3(g)] insofar as the appearance, value, marketability, or consumer acceptability is concerned and does not relate to any use of the ingredient as a color additive [21 CFR 70.3(f)].

Table 5 Intended Uses of lycopene and Maximum Use Levels*

Food Category	Proposed Food-Use	Use-Levels for Lycopene (ppm)
Baked Goods and Baking Mixes	Nutrition Bars	50
	Crackers	30
Beverages and Beverage Bases	Meal Replacements	25
Breakfast Cereals	Ready-to-Eat Cereals	50
Cheeses	Processed Cheese Spread	5
Condiments and Relishes	Sauces, Seasonings, Relishes, and Pickles	50
Confections and Frostings	Decorations, Filings, and Icings	25
Fats and Oils	Table Fat Spreads	5
	Low Fat Salad Dressings	20
Frozen Dairy Desserts and Mixes	Frozen Dairy Desserts	25
Gelatins, Puddings, and Fillings	Gelatin Desserts, Puddings, and Custards	25
Gravies and Sauces	Tomato-Based, Gravies, and Specialty Sauces	50
Hard Candy	Hard Candy	25
Milk Products	Dairy-Based Fruit Drinks	50
	Milk-Based Meal Replacements	25
	Cultured Dairy Drinks	20
Plant Protein Products	Meat Substitutes	50

Processed Fruits and Fruit Juices	Energy, Sport, and Isotonic Drinks	25
	Fruit-Flavored Drinks	25
	Fruit Juice	25
	Nectars	25
Snack Foods	Salty Snacks	30
Soft Candy	Jelly Products	25
Soups and Soup Mixes	Prepared and Condensed Soups	7
	Dry Soup Mixes	575

*Adapted from GRN 173, page 50/50.

* These food-uses represent non-standardized food products; however, in order to obtain a conservative intake estimate, surrogate codes for the standardized food products were chosen.

3.2. Estimated Daily Intake from the Proposed Uses

Using data from the National Health and Nutrition Examination Survey (NHANES) conducted by the National Center for Health Statistics from 2017 to 2020 cycle, an assessment was conducted on the expected intake of lycopene under proposed food uses, as summarized in Table 6. The NHANES data was used to estimate the mean and 90th percentile daily intakes of lycopene among the U.S. population aged two years and older, and the intake estimates are based on those who consumed one or more of the target foods on the day of the survey. A conservative approach was adopted, assuming the addition of lycopene at the maximum proposed levels in each food category. The estimation is conservative, assuming that the levels of lycopene added to each proposed food category is at the maximum proposed level.

Statistical analysis and data management were performed using the DaDiet software (Dazult Ltd., 2018), a web-based tool designed to estimate exposure to nutrients and substances added to food, including contaminants, food additives, and novel ingredients. The main input components for this software are concentration levels (usage levels) and food consumption data. These datasets were integrated within the software to provide accurate and efficient exposure assessments. A summary of the daily intake estimates for lycopene under proposed use in selected food categories is provided in Table 6 on an absolute basis (mg/person/day) and in Table 7 on a body weight basis (mg/kg bw/day).

In the U.S. population over 2 years old, the mean intake and the 90th percentile intake of lycopene, when added to designated foods, were determined to be 5.93 mg/person/day and 13.11 mg/person/day, respectively. Male teenagers exhibited the highest mean and 90th percentile intakes of lycopene, at 7.58 mg/person/day and 15.80 mg/person/day, respectively. On a body weight basis, the mean and 90th percentile intakes of lycopene for the U.S. population aged two years and older were established at 0.099 mg/kg bw/day and 0.23 mg/kg bw/day, respectively. Among specific population groups, male children were identified as having the highest mean and 90th percentile intakes (based on body weight), at 0.29 mg/kg bw/day and 0.59 mg/kg bw/day, respectively.

Table 6. Summary of the Estimated Daily Intake of Lycopene from Proposed Food Uses (mg/day) Based on 2017-2020 NHANES Data

Population Group	Per Capita Intake (mg/day)		Per Consumer Intake (mg/day)			
	Mean	90 th percentile	%	n	Mean	90 th percentile
Children, Male (2-5 yr)	4.883	10.479	99.3%	396	4.916	10.69
Children, Female (2-5 yr)	4.887	9.524	99.5%	400	4.889	9.524
Children, Male (6-11 yr)	7.299	15.631	99.0%	598	7.372	15.631

Children, Female (6-11 yr)	6.698	12.881	97.9%	593	6.729	12.888
Teenagers, Male (12-18 yr)	7.447	15.769	98.3%	651	7.577	15.801
Teenagers, Female (12-18 yr)	6.867	14.761	98.1%	642	7.014	15.069
Adults, Male 19+ yr	6.892	15.139	98.3%	3161	7.008	15.502
Adults, Female 19+ yr	4.432	10.475	98.3%	3458	4.518	10.579
Population 2+ yr	5.834	13.021	98.4%	9899	5.931	13.112

n = sample size; yr=Year; *Calculated by using DaDiet, a Dietary Intake Evaluation Tool by Dazult.

**Table 7. Summary of the Estimated Daily Intake of Lycopene from Proposed Food Uses (mg/kg bw/d)
Based on 2017-2020 NHANES Data**

Population Group	Per Capita Intake (mg/kg bw/day)		Per Consumer Intake (mg/kg bw/day)			
	Mean	90 th percentile	%	n	Mean	90 th percentile
Children, Male (2-5 yr)	0.283	0.585	99.5%	384	0.285	0.589
Children, Female (2-5 yr)	0.297	0.55	99.5%	396	0.298	0.55
Children, Male (6-11 yr)	0.229	0.555	99.0%	596	0.231	0.559
Children, Female (6-11 yr)	0.203	0.398	97.9%	590	0.204	0.398
Teenagers, Male (12-18 yr)	0.119	0.285	98.3%	643	0.121	0.287
Teenagers, Female (12-18 yr)	0.117	0.255	98.1%	639	0.12	0.259
Adults, Male 19+ yr	0.0797	0.177	98.3%	3133	0.081	0.179
Adults, Female 19+ yr	0.0607	0.138	98.3%	3420	0.0618	0.138
Population 2+ yr	0.0971	0.231	98.4%	9801	0.0987	0.233

bw = body weight; n = sample size; yr=Year.*Calculated by using DaDiet, a Dietary Intake Evaluation Tool by Dazult.

3.3. Natural Sources and Intake

The primary source of natural lycopene in the human diet is tomatoes and products derived from it. Additionally, fruits like pink grapefruit, water melon, rosehip, papaya and guava are also sources of lycopene (Nguyen and Schwartz, 1999). Lycopene from tomatoes, synthetic lycopene and lycopene synthesized by the fungus *B. trispora* are permitted for uses in food. In the US, over 80% of dietary lycopene intake is derived from processed tomato products such as ketchup, tomato juice, spaghetti sauce, and pizza sauce (Clinton, 1998). The amount of lycopene present in processed foods is often much higher as compared to that found in fresh foods, as processing often involves concentration via water loss. Tomatoes and tomato-based products are the major and the best sources of lycopene in the human diet. A list of tomato products with lycopene is given in Table 8, while main dietary sources of lycopene are given in Table 9.

As mentioned earlier, lycopene is present in human diet, primarily in dark red fruits and vegetables, such as tomatoes, watermelons, and sweet red peppers. Thus it has been ingested for hundreds of years from foods without any harmful health effects. The available information suggest that a 130 g serving of fresh tomatoes contains 4-10 mg of lycopene, while ketchup contains 3.3 mg per tablespoon. The mean level of lycopene consumed from food is estimated as 8 mg/day. Lycopene is available as dietary supplement (2,446 labels found) and it is most often used by adults in doses of 15-45 mg by mouth daily for up to 6 months. As per 21 CFR 73.585, tomato lycopene extract; tomato lycopene concentrate are approved as color additive. Tomato lycopene is included in EAFUS list, it is also recognized by Flavor and Extract Manufacturers Association as GRAS ingredient (FEMA No.: 4110).

Table 8. Lycopene Content of Tomato Foods

Food	Lycopene content (mg/100 g)
Tomatoes, raw	0.9 - 4.2
Tomatoes, cooked	3.7 - 4.4
Tomato sauce	7.3 - 18.0
Tomato paste	5.4 - 55.5
Tomato soup (condensed)	8.0 - 10.9
Tomato juice	5.0 - 11.6

*Shi and Le Maguer (2000)

As carotenoids are considered to be nonessential micronutrients, there is no consensus on recommended daily dose of lycopene. Depending on the study population, dietary intake of lycopene varies greatly. In Italy, the average total consumption of carotenoids is reported as 14.3 mg/day. Despite significant individual disparities, the average consumption of lycopene varies between 5 and 7 mg/day in the western world (Petyaev, 2016). In Italian diet, lycopene makes up the largest proportion of carotenoids, with an average intake of 7.4 mg/day (Lucarini et al., 2006). The average daily intake of lycopene in the United States ranges from 6.6 - 10.5 mg/day for men and from 5.7 - 10.4 mg/day for women, 3.8 mg/day in Australia, 1.1 mg/day in the United Kingdom. In the industrialized European countries, daily intake of lycopene varies from 0.7 mg (Finland) to 1.3 mg (Germany) (Kaur et al., 2017). Shao and Hathcock (2006) reported that higher intake of lycopene at daily dosage up to 100 mg has no side effects in volunteers. For lycopene, there is no evidence that when consumed in foods there is a cumulative effect in the diet that is unsafe.

Table 9. Main Dietary Sources of Lycopene*

Food	Lycopene content (mg/100 g)
Tomato (<i>Solanum pimpinellifolium</i> L.)	0.594–43.09
Tomato (<i>Solanum lycopersicon</i> L.)	1.60–18.46
Pink guava (<i>Psidium guajava</i> L.)	5.43–7.02
Watermelon (<i>Citrullus lanatus</i> (Thumb.) Mansf.)	2.45–7.30
Pink grapefruit (<i>Citrus paradisi</i>)	3.32–3.36
Papaya (<i>Carica papaya</i> L.)	2.54–3.72
Persimmon (<i>Diospyros kaki</i> L.)	0.158–0.359
Mango (<i>Mangifera indica</i> L.)	0.072–0.082
Wild cherry (<i>Prunus avium</i> L.)	0.010

*Adapted from Camara et al. (2013)

4. PART IV- SELF LIMITING LEVELS OF USE

No known self-limiting levels of use are associated with the use of lycopene preparation.

5. PART V- EXPERIENCE BASED ON COMMON USE IN FOODS BEFORE 1958

The statutory basis for the conclusion of GRAS status of lycopene preparation in this document is not based on common use in food before 1958. The present GRAS assessment for the use of lycopene preparation as a food ingredient is based on scientific procedures.

6. PART VI- NARRATIVE

The safety determination of lycopene produced by genetically engineered *S. cerevisiae* strain JZL03 is based on the totality of the available evidence, including animal experimental studies, *in vitro* studies, human studies of naturally occurring and synthetic lycopene, as well as uses of lycopene as a food ingredient. As lycopene preparation ($\geq 96\%$ pure), the subject of this GRAS determination, produced by *S. cerevisiae*, is substantially similar to lycopene naturally present in food, derived from tomato and synthetic, the available safety database that exists for lycopene and its preparation can be applied to establish the safety of lycopene preparation. The FDA raised no questions in response to GRAS notices submitted for use of lycopene derived from *Blakeslea trispora* (GRN 173), tomato lycopene (GRN 185; GRN 156), and synthetic lycopene (GRN 119) for use in numerous foods. Similarly, European Food Safety Authority (EFSA) has extensively evaluated the safety in use of lycopene as a food color, and the included sources of lycopene were tomatoes, synthetic lycopene and lycopene synthesized by the fungus *B. trispora*. Studies of acute oral toxicity, subchronic toxicity, chronic toxicity, reproductive and developmental toxicity, genotoxicity and carcinogenicity were included in the EFSA evaluation. In an expanded abstract on safety of lycopene, Trumbo (2005) from FDA concluded that dietary lycopene intake at levels typically consumed in the US appears to be safe among the general healthy population.

For the present GRAS assessment, efforts have been made to present both the data supporting safety of lycopene as well as any data on potential adverse effects. An attempt has been made to interpret these findings from relevant studies as they relate to the present GRAS assessment. The assessment of efficacy studies is limited to a review of the results related to safety and tolerability. For this assessment, a comprehensive search of the scientific literature was conducted through December 2024 to identify publications on the safety of lycopene. In the following sections, relevant biological and toxicological studies on lycopene are described that provide support for the conclusions reached in this determination.

6.1. Data Pertaining to Safety

Lycopene has been the subject of four GRAS determinations evaluated by FDA (GRN 119 for synthetic lycopene; GRN 156 tomato; GRN 173 for lycopene from *Blakeslea trispora*; GRN 185 for concentrated tomato Lycopene extract). Comparison of lycopene from *Saccharomyces cerevisiae* with other sources that are marketed is provided in Table 10. For synthetic lycopene that is subject of GRN 119, the available information from SCF (1999) is included. This information shows that the purity of lycopene derived from *S. cerevisiae* is similar to three GRAS notices (GRN 119, GRN 173, GRN 185). This comparison also suggest that the safety data and information from these three GRAS notices is applicable to the subject of present GRAS. The EFSA panel established the Acceptable Daily Intake (ADI) at 0.5 mg/kg bw/day, based on the No Observed Effect Levels (NOAEL) for synthetic lycopene formulation in various toxicity studies.

Shao and Hathcock (2006) systematically evaluated the research designs and data from multiple studies that provide a basis for risk assessment and the usual tolerable Upper Level of Intake (UL) derived from it if the newer methods described as the Observed Safe Level (OSL) or Highest Observed Intake (HOI) are utilized. The OSL risk assessment method indicates that the evidence of safety is strong at intakes up to 75 mg/day for lycopene, and these levels are identified as the respective OSL. Although much higher levels have been tested without adverse effects and may be safe, the data for intakes above these levels are not sufficient for a confident conclusion of long-term safety.

Table 10. Comparison of Lycopene from *Saccharomyces cerevisiae* with other sources that are marketed

Parameters	<i>S. cerevisiae</i>	Synthetic*	Tomatoes (GRN 156)	<i>B. trispora</i> (GRN 173)	Tomato Lycopene (GRN 185)
Purity	>96%	>96%	>5%	>95%	>90%
Impurities, other pigments	-	0.3% C ₂₅ aldehyde	Other pigments, oils, fats, waxes and natural flavors	Other carotenoids	-
All-trans isomer	>94%	>70%	94-96%	95-99%	94%
5-cis isomer	<6%	<25%	3-5%	1-5%	1.7% (cis)
9-cis isomer		<1%	0-1%		
13-cis isomer		<1%	1%		
Other cis isomer		<3%	1%		3.1%

*Synthetic Lycopene as a coloring matter for use in food (SCF, 1999). It should be noted that synthetic lycopene is GRAS as per GRN 119.

6.2.1. Absorption, Distribution, Metabolism and Excretion of Lycopene

The available information suggest that fat soluble carotenoids, such as lycopene, are absorbed in the intestine and transported in chylomicrons to the blood stream via the lymphatics. Several studies related to the absorption, distribution and excretion of lycopene from various sources, including natural lycopene from tomato concentrate and synthetic lycopene, have been performed. The findings from animal and human intervention studies with tomato extract as lycopene source and with synthetic lycopene indicate that lycopene from these sources is bioavailable. The bioavailability of synthetic, formulated lycopene is comparable to tomato-based lycopene (Cohn et al., 2004; Hoppe, 2003). It is expected that lycopene from *S. cerevisiae* when used in food products of comparable composition will be bioavailable to a similar extent as lycopene from tomatoes.

Bioavailability of lycopene can be affected by several factors, including food processing and dietary composition (Story et al., 2010). Lycopene occurs in several forms in fresh plant foods, including carotenoid-protein complexes in chloroplasts or in crystalline form inside chromoplasts. Thermal processing generally improves lycopene bioavailability by disrupting cellular membranes, which allows lycopene to be released from the tissue matrix. The available evidence suggest that lycopene from thermally processed tomato products is more bioavailable than lycopene from fresh tomatoes. The absolute amount of lycopene absorbed does not appear to vary greatly with dose. Inter-individual differences, not dose, has the greatest impact on the amount of lycopene absorbed. Lycopene bioavailability is greatly affected by dietary composition. Given that lycopene is a lipid-soluble compound, consuming it with fat increases its bioavailability. *In vivo* studies indicate that the cis-isomers of lycopene appear to be more bioavailable than the all-trans isomer. Diwadkar-Navsariwala et al. (2003) reported that the lack of toxicity of high doses of lycopene may be due to the fact that lycopene absorption has an upper limit of saturated concentration and is not absorbed when the dose exceeds 60 mg.

Following ingestion, lycopene must first be released from the food matrix before it is incorporated into mixed micelles (Story et al., 2010). The available information indicate that lycopene was absorbed by the same route as dietary lipids, i.e., passive diffusion. Subsequent studies reported that lycopene absorption can be facilitated by a cholesterol membrane transporter known as scavenger receptor class B type I. Once inside the enterocyte, lycopene is packaged with other dietary lipids into chylomicrons that are transported across the basolateral membrane and make their way into the lymphatic system, which eventually releases chylomicrons into the blood. Competition with other carotenoids or cholesterol may also influence lycopene absorption.

The available studies suggest that lycopene is stored in human organs at varying degrees (Kun et al., 2006). Lycopene is found in the highest concentrations in the liver, testes, adrenal glands, and adipose tissues, while it is found in lower concentrations in the kidney, ovary, lung, and prostate. The available data indicate that in rats lycopene is absorbed in systemic circulation and is distributed to the tissues in similar pattern as seen in humans. Tissue levels of lycopene in rats are comparable with that in humans. This also suggest that rat can be considered as a useful and appropriate animal model for lycopene studies.

In a recent review article on lycopene digestion, absorption, metabolism, and excretion, Arballo et al. (2021) reported that majority of lycopene is cleaved eccentrically by the enzymes β -carotene oxygenase 2 (BCO2), yet further research is needed to probe the enzymatic cleavage activity at the tissue level. Additionally, results indicate that single nucleotide polymorphisms and dietary fat influence lycopene absorption and thus modify its health effects. These investigators noted, high inter-individual variability in lycopene levels in humans, attributed in part to food preparation procedures, differential bioavailability depending on the presence of co-consumed dietary fat, and differences in genetic polymorphisms associated with lycopene absorption and metabolism.

6.2.2. Specific Corroborative Studies

In a series of unpublished toxicity studies, potential adverse effects of lycopene preparation, subject of present GRAS, containing >96% lycopene were investigated. These safety studies of lycopene preparation included, acute toxicity, and mutagenicity and genotoxicity (Ames test, *in vitro* mammalian chromosome aberration test and *in vivo* mammalian micronucleus test). All studies were conducted according to the China National Food Safety Standards (known as Guo Jia Biao Zhun, or GB Standards). Translated copy of the study report was provided by Hesheng Technology.

6.2.2.1. Acute Toxicity of Lycopene Preparation

In an unpublished acute oral toxicity study, conducted as per National Standards of the People's Republic of China: GB 15193.3-2014 National Food Safety Standard - Acute Oral Toxicity Test, oral toxicity of lycopene preparation was evaluated in SPF healthy adult SD rats. For this study, lycopene preparation from Hesheng Technology was used. In this study, 20 (10/sex) adult Specific Pathogen-Free (SPF) Sprague-Dawley rats were used. The test was carried out as a limit test and only one dose group was included. Purified water was used as the solvent and the test article was prepared immediately before use. Lycopene preparation was administered by a single oral gavage at a dose of 10 g/kg bw. Onset time, extent and duration of changes in animals were comprehensively observed and documented to evaluate possible target organs of toxicity. Observations were made on the skin, hair coat, eyes, mucous membranes, and respiratory, urinary, digestive, and nervous systems.

Rats were observed for toxicity immediately after administration, and the onset time and end time of toxic symptom as well as death were recorded. The observation period was 2 weeks. Euthanasia was performed on animals that were moribund, suffering from severe pain, and still alive at the end of test. Oral gavage administration of lycopene preparation did not result in significant toxicity or death in the animals during the 14-day observation period. Weight measurements on days 7 and 14 post-dose showed weight gain in surviving animals. The animals were euthanized at the end of the test, and no abnormalities were found in gross necropsy. The LD₅₀ for the acute oral toxicity test of lycopene preparation is >10 g/kg bw. The findings from this study suggest that lycopene preparation was actually non-toxic in the acute oral toxicity test in rats.

6.2.2.2. Mutagenicity and Genotoxicity Studies of Lycopene Preparation

6.2.2.2.1. Bacterial Reverse Mutation Test of Lycopene Preparation

This test was conducted as per National Standards of the People's Republic of China: Bacterial Reversion Mutation GB15193.4-2014. The bacterial reverse mutation was conducted using five standard histidine-deficient *Salmonella* Typhimurium mutant strains (TA97a, TA98, TA100, TA102, and TA1535). Rat liver microsomal enzymes (S9 mixture, 0.5 ml/dish) induced by a combination of phenobarbital sodium and β -naphthoflavone were used as the *in vitro* metabolic activation system. Five dose groups (5000, 2000, 1000, 500, and 200 μ g/dish) were tested in the pilot test. The results showed no significant difference in background bacterial lawn between each dose group and the untreated control group, indicating that the test article did not induce bacterial toxicity to the test strains at doses of 5000 μ g/dish and below. Based on the pilot test results, 5 dose groups (5000, 1582, 500, 158, and 50 μ g/dish) were set up for the main test, along with an untreated control group, a solvent control group (DMSO), and a positive mutagen control group.

In the formal and validation tests, lycopene preparation precipitation was not observed after the addition of test solutions in ≤ 5000 μ g/dish groups with or without metabolic activation. There was no significant difference in the background lawn of each test strain between each dose group and control group, indicating that the test article had no bacterial toxicity at doses $\leq 5,000$ μ g/dish. The number of revertants for each test strain in each dose group did not exceed 2 times as compared with the untreated control group with or without metabolic activation. No dose-response relationship and no test points with reproducible positive results were observed. The mean number of revertants for each test strain in the positive control group was over 2 times that in the untreated control group, showing a significant positive reaction. These findings indicate that the lycopene preparation did not induce an increase in the number of reversion colonies for the five strains. Lycopene preparation is negative for the bacterial reverse mutation test.

6.2.2.2.2. In Vitro Mammalian Chromosome Aberration Test of Lycopene Preparation

This assay was conducted in accordance with the People's Republic of China: National Food Safety Standard - *In Vitro* Mammalian Chromosome Aberration Test GB 15193.23-2014. For this test, the Chinese hamster lung (CHL) cell line was selected. The rat liver microsomal enzyme (10% S9 mixture) induced by phenobarbital sodium and β -naphthoflavone served as the *in vitro* metabolic activation system. DMSO was used as the vehicle. The maximum available solubility of the sample was approximately 160 mg/ml. Cells were treated with lycopene preparation at concentrations (doses) of 100, 200, 400, and 800 μ g/ml for 4 hours without metabolic activation, with purified water as the vehicle. The results showed that the test article did not affect cell count. Cell viability in the 100, 200, 400, and 800 μ g/ml groups at the end of culture

were 96.12%, 103.88%, 75.19%, and 58.14%, respectively. Additionally, cells were also treated with the same concentration of lycopene preparation and without metabolic activation, for 24 hours. The results showed that the lycopene preparation did not affect cell count. Cell viability in the 100, 200, 400, and 800 µg/ml groups at the end of culture were 102.20%, 95.60%, 71.98%, and 56.59%, respectively. Based on the results of the pilot experiment, three dose groups (200, 400, and 800 µg/ml) were set for the formal test along with a vehicle control group (DMSO) and two positive control groups [without S9 mixture: methyl methanesulfonate (MMS, 20 µg) /ml), with S9 mixture: cyclophosphamide (CP, 10 µg/ml)].

The results of three test conditions (without or with S9 for 4 hours, and without S9 for 24 hours), lycopene preparation solution did not affect the cell observations and counts in the 200, 400, and 800 µg/ml groups. After 4 hours of treatment with and without S9, the chromosome aberration rates of CHL cells were 2.0% and 2.0% in the vehicle control group, and 1%, 2%, 1%, and 2%, 2%, 0% in the lycopene dose groups, respectively. There was no significant difference in the chromosome aberration rate between the lycopene dose groups and vehicle control group. The chromosome aberration rates of CHL cells in the positive control groups (MMS and CP) were 9% and 9%, respectively, which was significantly higher than those in the vehicle control group. As regards after 24 hours of treatment with and without S9, the chromosome aberration rate of CHL cells was 0% in the vehicle control group, and 2%, 0% and 2% in the lycopene dose groups, respectively. There was no significant difference in the chromosome aberration rate between the lycopene dose groups and vehicle control group. The chromosome aberration rate of CHL cells in the positive control group (MMS) was 9%, which was significantly higher than that in the vehicle control group. The findings from this study demonstrate that lycopene preparation does not cause chromosome aberration in CHL cells under the test conditions.

6.2.2.2.3. *In Vivo* Mammalian Micronucleus Test of Lycopene Preparation

In a study in mice effects of lycopene preparation on micronucleation in immature erythrocytes was investigated. The study was conducted as per National Standards of the People's Republic of China: GB 15193.5-2014 National Food Safety Standard, effects on incidence of micronucleation in immature erythrocytes of mice. For this study, SPF Kunming mice (25/sex; weighing 25-30 g) were used. For this study, three dose groups for treatment with lycopene preparation were set, i.e., low dose (1250 mg/kg bw), intermediate dose (2500 mg/kg bw), and high dose (5000 mg/kg bw). Animals in the positive control group were administered cyclophosphamide by oral gavage at 40 mg/kg bw. The test article (lycopene preparation) was administered via gavage twice within 30 hours, that is, a second dose was administered 24 hours after the first dose by oral gavage.

Animals were observed once a day for changes in the skin, hair, eyes, mucous membranes, respiratory system, circulatory system, nervous system, limb activities and behavior. For sampling, the animals were euthanized by cervical dislocation 6 hours after the second dose. Once blood and muscles were removed, the sternum was cut transversely to expose the medullary cavity. The sternal bone marrow was squeezed out using a small hemostat, dropped onto the calf serum at one end of the slide and mixed thoroughly. The smear was prepared using a push technique.

The smear was air dried after spreading, the slide specimen was fixed in methanol for 5 to 10 min and air dried, stained with Giemsa, and microscopically examined. Areas with intact, evenly dispersed, and properly stained cells were selected under low magnification and then observed under oil immersion microscopy. A total of 2,000 polychromatic erythrocytes (PCEs)

were counted per animal. The number of micronucleated PCEs was also counted to calculate the incidence of micronucleation by the following formula: Incidence of micronucleation (‰) = Number of micronucleated PCEs/total number of PCEs x 1000. The number of immature erythrocytes among 200 erythrocytes (PCE+NCE) was counted for each animal to calculate the percent of immature erythrocytes as follows: Percent of immature erythrocytes (%) = PCE/(PCE+NCE) x100.

During the treatment period, no animal death or significant abnormal changes were observed in the activities, feed intake, water intake, and defecation of animals in each group. The percent of PCE in each dose group of the test article and the positive control group was not lower than 20% of that in the solvent control group, and the difference was not statistically significant. This suggest that the test article, lycopene preparation, has no inhibitory effect on bone marrow cells of female and male Kunming mice.

The incidence rates of PCE micronuclei in the bone marrow of female and male mice in the solvent control group were 1.3‰ and 1.4‰, respectively. The incidence of PCE micronuclei in the bone marrow of female and male mice in each dose group of the samples submitted for inspection ranged from 1.2‰ to 1.6‰. Compared with the solvent control group, the difference was not statistically significant; the solvent control group and each Micronucleus incidence rates across dose groups were within the range of the laboratory's historical background control data. The incidence rates of PCE micronucleation in the bone marrow of female and male mice in the positive control group were 33.5‰ and 29.8‰, respectively, which were significantly higher than that in the control group. The incidence of micronuclei in the positive control group was within the range of laboratory historical background control data. The results showed that the lycopene preparation did not lead to increased incidence of PCE micronucleation in the mouse bone marrow.

6.2.2.3. Summary of Specific Toxicity Studies of Lycopene Preparation

In summary, the product specific studies did not reveal any adverse effects of lycopene preparation. In acute toxicity in rats, the LD₅₀ of lycopene preparation was >10 g/kg bw indicating that lycopene preparation is practically safe. In mutagenicity and genotoxicity studies as evaluated by Ames assay, *in vitro* mammalian chromosomal assay and *in vivo* mammalian micronucleus assay was negative. Although these studies are not yet published, the findings from all these studies corroborate the safety of lycopene preparation, the subject of this GRAS determination.

6.2.3. Animal Toxicity Studies

In multiple animal toxicity studies, potential adverse effects of lycopene has been investigated. An overview of the available animal toxicity studies of lycopene derived from different sources is provided in Table 11. In an acute toxicity study, the oral LD₅₀ value of lycopene in mice was assessed to be higher than 3000 mg/kg bw (Milani et al., 1970). The source of lycopene used in this study is not specified in the publication.

In a subchronic toxicity study, Jonker et al., (2003) investigated the toxicity of lycopene derived from a fungal biomass (*B. trispora*). In this study, lycopene, as a suspension in sunflower oil (20% w/w) was fed at dietary concentrations of 0, 0.25, 0.50, and 1.0% to groups of Wistar rats (20/sex/group) for a period of 90 days. Lycopene intake was calculated to be 0, 145, 291, and 586 mg/kg bw/day in control through high-dose males and 0, 156, 312, and 616 mg/kg bw/day in control through high-dose females. The results from this study do not provide any evidence of toxicity of lycopene at dietary levels up to 1.0% as demonstrated by the findings of clinical

observations, neurobehavioral observations, motor activity assessment, body weight and food consumption measurements, ophthalmoscopic examinations, hematology, clinical chemistry, urinalysis, organ weights, gross pathology, or histopathology. The NOEL from this 90-day oral toxicity study with lycopene extracted from *B. trispora* was determined as 1.0% in the diet (586 mg lycopene/kg bw/day in males and 616 mg lycopene/kg bw/day in female), the highest dietary concentration tested.

In the published literature, multiple animal toxicity studies of lycopene have been reported (Table 11). In these studies effects of lycopene derived from sources such as tomato, *Blakeslea trispora* and synthetic have been used. These studies included short-term, subchronic, genotoxic and reproductive toxicity of lycopene. In three subchronic toxicity studies, no significant toxicological findings were reported in male and female rats following 13 weeks (Mellert et al., 2002) 14 weeks (McClain and Bausch, 2003) following oral exposure to synthetic lycopene and in male and female Sprague Dawley rats following 13 weeks exposure to tomato oleoresin extract (East, 1995). In all these studies NOAEL corresponded to the highest dose tested i.e., 300 mg lycopene/kg bw/day (Mellert et al., 2002), 500 mg/kg bw/day (McClain and Bausch, 2003) and 270 mg lycopene/kg bw/day (East, 1995).

In the thirteen-week studies by Mellert et al. (2002), effects of synthetic lycopene 10 CWDO and Lyco VitO 10% were investigated. Lycopene 10 CWDO was administered to male and female Wistar rats (10 rats/sex/group) at doses of 0, 500, 1500, or 3000 mg/kg bw/day. Lyco VitO was administered at doses of 0 or 3000 mg/kg bw/day. Red discoloration was seen in the jejunum and cecum of the treatment groups, but there were no histopathological changes. The NOAEL for both products was 3000 mg/kg bw/day, equivalent to 324 mg synthetic lycopene/kg bw/day.

In several additional studies in mice, rats and dogs that lasted for 2 weeks to 10 months, effects of oral exposure to lycopene (naturally occurring and synthetic) at doses up to 1000 mg/kg bw/day were investigated. These studies are summarized in Table 11. Regardless the source of lycopene, the totality of the evidence suggest that lycopene is well tolerated and there were no adverse effects on mortality, body weight gains, food consumption or clinical observations. With the exception of few studies where discolored tails in few animals was noted, no changes in coat appearance was noted at doses of dietary exposure up to 60 mg lycopene/kg bw/day (Zhao et al., 1998). In other studies, treatment related effects on major organs were limited to slight pigment accumulations in the liver and kidney in small groups of rats fed 10 to 20 mg lycopene/kg bw/day for 200 days and in one dog administered 100 mg lycopene/kg bw/day for 192 days (Zbinden and Studer, 1958).

In a publication on safety of synthetic lycopene as a water-dispersible beadlet formulation containing antioxidants, McClain and Bausch (2003) described acute and subchronic safety studies, reproductive studies, genotoxicity studies, metabolic studies, and exploratory studies on the hepatic uptake of lycopene. These investigators reported that lycopene has a low order of acute toxicity and no significant toxicity has been observed in rats treated with lycopene beadlet formulations in the diet at doses of up to 500 mg/kg bw/day for 14 weeks or 1000 mg/kg bw/day for 4 weeks. No teratogenic effects were noted in a rat two-generation study (1000 ppm in the diet) or in a teratology study in rats with 1000 mg/kg bw/day lycopene as beadlet formulations. It was noted that lycopene accumulates in hepatocytes and to a lesser extent in spleen. In short-term studies with synthetic lycopene, as a beadlet formulation, and natural source lycopene, as tomato concentrate, the accumulation of lycopene in the liver and the presence of pigment deposits in the

hepatocytes were similar and neither was associated with any histopathological changes. The pigment deposits in hepatocytes are no longer present after approximately 13 weeks of depletion, demonstrating reversibility for this effect.

In a series of studies, Matulka et al. (2004) determined the oral, dermal effects of lycopene as a 6% solution in tomato oleoresins. Lycopene preparation was evaluated for toxicological effects. In acute toxicity study, oral LD50 (50% lethal dose) in rats was greater than 5000 mg/kg bw. In acute dermal toxicity study, no toxicity was noted at 2000 mg/kg bw. Lycopene preparation lacked dermal irritation in the rabbit model, but was found to have moderate eye-irritant capabilities. Lycopene preparation tested at 5% (w/w) in petroleum jelly was a moderate sensitizer in the guinea pig model. These results demonstrate the inability of lycopene (6%) preparation to produce oral, or dermal toxicity in animal models at doses greater than 300 times the normal human consumption of lycopene.

In subchronic toxicity (13-week) study by Matulka et al. (2004), male and female CD rats (Charles River) (20 rats/sex/group) were orally administered 0, 45, 450, or 4500 mg/kg bw/day of a tomato extract containing an average of 5.8% lycopene (Matulka et al., 2004). Other than an increased incidence of orange-stained feces, there were no treatment-related effects. In this study, analysis of blood for lycopene concentrations after 13 weeks of treatment found that plasma levels of lycopene were similar in males given 450 or 4500 mg/kg bw/day, although a lower level was found in males given 45 mg/kg bw/day. These findings indicate that saturation occurs at a level between 45 mg/kg bw/day and 450 mg/kg bw/day. Under the test conditions, the NOAEL was 4500 mg/kg bw/day, 270 mg/kg bw/day as lycopene.

Christian et al. (2003) investigated potential oral developmental toxicity in rats and rabbits. In these experiments, each study included 3 control groups (water and matrix for Lycopene 10 CWD or LycoVit 10%), 3 Lycopene 10 CWD groups [500, 1500 and 3000 (rats)/2000 (rabbits) mg/kg bw/day] and 1 LycoVit 10% group [3000 (rats)/2000 (rabbits) mg/kg bw/day]. The highest dose used were maximum achievable concentrations and dosage volumes of the highly viscous test material suspensions. Animals were administered on gestation days (GDs) 6 through 19 (rats) or GDs 6 through 28 (rabbits). The animals were monitored for mortality, body weight, feed consumption, necropsy observations [GD 20 (rats)/GD 29 (rabbits)], uterine contents and fetal viability, gender, body weight and morphology (skeletons double-stained). Feed consumption and weight gain were essentially unaffected in rats and rabbits, despite intubation problems in both species and reduced gastrointestinal motility and mortality in rabbits attributable to the physical properties of the gels. In this study, lycopene 10 CWD or LycoVit 10% administration did not reveal direct maternal or developmental toxicity in rats or rabbits at dosages as high as 3000 or 2000 mg/kg bw/day, respectively. Expressed as lycopene, the NOAEL's are 324 mg/kg bw/day and 216 mg/kg bw/day, respectively.

Table 11. Summary of Animal Toxicity Studies of Lycopene*

Species	Duration	Dose (mg/kg bw/day)	Safety-related findings	Reference
Male and female Wistar rats	90 days	0.25, 0.5 or 1.0% (145 – 616 mg/kg bw/day) (dietary)	No significant toxicological effects	Jonker et al., 2003
Male and female Wistar albino	28 days	0.1, 0.3, or 1.0% (87 – 906 mg/kg bw/day) (dietary)	Lycopene versus control: Decreased prothrombin time and mean corpuscular volume (high-dose male	GRN 173

rats (20/sex/group)			rats). No significant toxicological effects	
Male and female Hanlbm Wistar rats (6/sex/group)	4 weeks	1000 (lycopene) + <0.01, 0.3, Or 2% apo-12'lycopenal (impurity) (oral)	Lycopene versus control: Red discoloration of the feces, brown-orange discoloration of the liver, brown-yellow fine granulated pigment deposits in the hepatocytes. No significant toxicological effects	McClain and Bausch, 2003
Male and female Hanlbm Wistar rats (26/sex/group)	14 weeks	50, 150 or 500 (oral)	No significant toxicological effects	McClain and Bausch, 2003
Male and female Wistar rats (10/sex/group)	13 weeks	50, 150, or 300 (500, 1500 or 3000 mg of 10% lycopene products) (gavage)	Lycopene versus control: Red discoloration of the feces, jejunum and cecum. No significant toxicological effects	Mellert et al., 2002
Male and female Fischer rats (20 control/10 per treatment)	10 weeks	2.4, 6.0, 12, 24, or 60 lycopene mg/kg bw/day (dietary)	Lycopene versus control: No toxic side effects. No adverse effects on weight gain, behavior, or coat appearance (brown discoloration of the tail in a few rats)	Zhao et al., 1998
SHN/Mei virgin mice (11 control/14 treated)	10 months	0.07 dietary	Lycopene versus control: No difference in body weight gain, no deleterious side effects detected	Nagasawa et al., 1995
Female SKH-Hr-1 hairless mice (30/group)	28 weeks	90 (dietary)	Lycopene versus control: No differences in mortality or body or liver weights	Black, 1998
Rat (10/sex/group)	100 days	1000 (dietary)	No adverse effects reported	Zbinden and Studer, 1958
Rat (numbers not specified)	200 days	10 to 20 (dietary)	Slight accumulation of pigments in the liver	Zbinden and Studer, 1958
Male hypercholesterolemia rats (numbers not specified)	28 days	7 groups: ranged from approximately 1.95 to 95 (dietary)	All lycopene groups (except 41 mg/kg body weight/day) versus control basal diet ↑'d serum cholesterol levels. Group receiving 9.6 mg lycopene/kg body weight/day versus control hypercholelemic diet: ↑'d liver cholesterol levels	Erdman and Lachance, 1973
Male and female Sprague-Dawley rats (20/sex/group)	13 weeks	45, 450, or 4500 natural tomato oleoresin extract (6% lycopene)	No significant toxicological effects	East, 1995
Male SPF Wistar rats (5/group)	15 days	45 (dietary)	Lycopene versus control: No difference in food intake, body weights, or absolute and relative	Gradelet et al., 1996

			liver weights. 60% ↓'d activity of liver enzyme nitrosodimethylamine N-demethylase (NDMAD)	
Male Wistar rats (8/group)	16 days	45 (dietary)	Lycopene versus control: No differences in food intake, body weight changes, or organ weights (small intestine, liver, lung, kidney). ↓'d activity of lung enzyme benzyloxyresorufin-O-dearylation (BROD)	Jewell and O'Brien, 1999
Female Wistar rats (4/group)	14 days	0, 1, 5, 50 or 100 (gavage)	Lycopene versus control: No differences in food intake, body weight changes or liver weights. ↑'d activity of liver enzymes benzyloxyresorufin-O-dealkylase (BROD) and ethoxyresorufin O-dealkylase (EROD)	Breinhold et al., 2000
Male F344 rats (22/group)	8 weeks	0, 0.5, 5.7, 57.5 (dietary)	Lycopene versus control: No difference in food intake	Boileau et al., 2000
1 dog	192 days	100 (capsules)	Pigment deposition in the liver and kidney	Zbinden and Studer, 1958

*Adapted from GRN 173

6.2.4. Mutagenicity and Genotoxicity Studies

In an unpublished study described in GRN 173, potential genotoxic effects of lycopene derived from *B. trispora* (20%) was investigated in bacterial (*Salmonella typhimurium* strains TA1535, TA1537, TA98, TA100 and *Escherichia coli* strain WP2 uvrA) and mammalian (human lymphocytes) test systems. In these studies no increase in revertant colony counts of any of the strains, either in the presence of or absence of S9 mix was noted. Similarly no statistically significant increase in the incidence of cells with chromosome was noted. These findings indicate that lycopene 20% showed no evidence of genotoxicity. Additionally, naturally occurring and synthetic formulations of lycopene were investigated in *in vitro* bacterial (*Salmonella typhimurium* and *E. coli* strains) and mammalian (mouse lymphoma cells and human lymphocytes) test systems, as well as *in vivo* with mice and rats. In the *in vitro* studies, consistently negative results were reported (He and Campbell, 1990; Rauscher et al., 1998; Aizawa et al., 2000; McClain and Bausch, 2003).

The degradation products of crystalline lycopene, as a result of being exposed to light and air, were shown to exhibit some mutagenic activity. Mutagenicity testing of formulated lycopene using mouse lymphoma cells did not reveal increase in mutant frequency with lycopene (McClain and Bausch, 2003). These investigators reported that unformulated pure crystalline lycopene and lycopene as a 10% beadlet formulation are not genotoxic as determined in a comprehensive battery of tests, however, improperly stored, unformulated crystalline lycopene can degrade to mutagenic products if exposed to light and air. There was no evidence of clastogenicity or DNA damage in mouse peripheral blood or rat hepatocytes, respectively following oral administration of synthetic lycopene formulation (McClain and Bausch, 2003). Oral exposure to naturally occurring lycopene

did not induce chromosomal damage in mouse bone marrow cells (Rauscher et al., 1998) or DNA damage in human lymphocytes (Pool-Zobel et al., 1997; Collins et al., 1998; Riso et al., 1999).

6.2.5. Human Studies with Lycopene

In a systematic review and meta-analyses, Tierney et al. (2020) determined the effects of consuming dietary and/or supplemental lycopene on cardiovascular risk factors. In this review and analysis, 43 studies were included. Lycopene interventions were highly variable (supplement with or without food, based as tomato juice/paste/raw product, or combined with olive oil), the dose ranged from 1.44 to 75 mg lycopene/day and was not reported in 11 of 43 included studies. The duration of effect studied was considerable and lasted from 30 min to 6 months. Studies reported conflicting findings for the effect of lycopene on cardiovascular risk factors. This was supported by meta-analyses where there were no significant differences between lycopene intervention and control groups for blood pressure and lipids (total cholesterol, LDL cholesterol, HDL cholesterol, and triglycerides). This was observed for overall groups and in subgroup analyses for individuals with elevated risk factor concentrations at baseline. Lycopene interventions for cardiovascular risk factors were highly variable across studies in both the dosage provided and the mode of delivery (supplement or food based). No adverse effects of lycopene were reported.

In another review, Shao and Hathcock (2006) reviewed the available human studies of lycopene. For this risk assessment, the inclusion criteria were: study duration; at least 1 week, lycopene dose; not less than 8 mg/day, and study design; only randomized, placebo-controlled intervention trials (Table 12). Studies that were uncontrolled and unblinded and studies investigating acute bioavailability, pharmacokinetics or post-prandial responses from single bolus doses were excluded. Studies that did not quantify the administered dose of lycopene (such as certain feeding studies) were also excluded. In total, 16 human safety studies were included, in which the highest lycopene dose was 150 mg/day (for 7 days) and the longest study duration was 20 weeks (the lycopene dose was 13.3 mg/day). None of the clinical trials found an adverse effect related to lycopene administration. Hence, no LOAEL or NOAEL could be established and Upper Limit (UL) could not be set.

In several clinical studies with lycopene supplementation, using tomato-based products or tomato based capsules, various endpoints were investigated. These studies did not reveal any abnormalities in body weights, full blood counts, liver function tests or immune function tests in subjects supplemented with lycopene at levels ranging from 0.5 mg/day for 4 weeks, 15 or 30 mg for 3 weeks, 47.1 mg/day for 8 weeks to 75 mg/day for 1 week, and these doses were generally well tolerated without reports of any illness or adverse biological effects (EFSA, 2008). However, also these studies have not been designed to evaluate the safety of lycopene.

Table 12. Human Clinical Studies of Lycopene*

Study population (n)	Dosage and study design	Duration (days)	Key observations	OSL considerations
Healthy adults n =20	75, 150 mg/day lycopene (as tomato oleoresin); randomized,	7 days	Significant 2, and 2.4-fold increase, respectively, in serum lycopene (to 661.9 and 793.8 nmol/L) vs.	250 mg, but relatively short duration and small sample size argue against use of this study for identification of an OSL

	controlled, crossover		placebo; no adverse effects reported	
Healthy adults n = 15	70 mg/day lycopene (as beadlets), 75 mg/day (as oleoresin and tomato juice, respectively); randomized, controlled, crossover	4 week (28 days)	Significant 41% increase in serum lycopene (to approx. 825 nmol/L) for all three treatments; significant 2-fold increase in buccal mucosa cell lycopene level for beadlet and oleoresin forms, respectively; no adverse effects reported	75 mg in healthy adults, substantial duration of exposure to multiple forms at a high dosage; based on the findings of the study above, this study is chosen to serve as the basis for the human OSL
Healthy elderly n = 29	47 mg/day lycopene (as tomato juice); randomized, controlled	8 week (56 days)	Significant 4-fold increase in serum lycopene (to 540 nmol/L); no adverse effects reported	47 mg, supports the OSL selected
Exercise-induced asthma patients n = 20	30 mg/day lycopene (as tomato oleoresin); randomized, controlled, crossover	1 week (7 days)	Significant 2-fold increase in serum lycopene (to 149 nmol/L); no adverse effects reported	30 mg, supports the OSL selected
Healthy adults n = 23	15 mg/d lycopene (as tomato oleoresin); randomized, controlled	26 days	Significant 44% increase in serum lycopene (to 1010 nmol/L); no adverse effects reported	15 mg, supports the OSL selected
Healthy men n = 52	15 mg/day lycopene (as tomato oleoresin); randomized, controlled	12 week (84 days)	Significant 1.9-fold increase in serum lycopene (to 1170 nmol/L); no adverse effects reported	15 mg, supports the OSL selected
Healthy female adults n = 10	45 mg/day lycopene as tomato juice; randomized, controlled	29 days	Significant 2.8-fold increase in serum lycopene (to 1842 nmol/L); no adverse effects reported	45 mg, supports the OSL selected
Healthy elderly n = 17	13.3 mg/day lycopene (as tomato oleoresin); randomized, controlled	12 week (84 days)	Significant 2.7-fold increase in serum lycopene (to 270 nmol/L); no adverse effects reported	13.3 mg, supports the OSL selected
Prostate cancer patients n = 15	30 mg/day lycopene (as tomato oleoresin);	3 week (21 days)	No significant changes for lycopene in serum or tumor tissues; no adverse effects observed	30 mg, supports the OSL selected

	randomized, controlled			
Healthy adults n = 93	13.3 mg/day lycopene (as tomato oleoresin) randomized, controlled	20 week (140 days)	Significant 2-fold increase in serum lycopene (to 1200 nmol/L); no adverse effects reported	13.3 mg, supports the OSL selected
Healthy adults n = 22	25 mg/day lycopene (as tomato paste or oleoresin, respectively); randomized, controlled	8 week (56 days)	Significant 2.7 and 2.5-fold increases in serum lycopene in the two treatment groups, respectively; significant 3.7-fold increase in buccal mucosa cell lycopene level; no adverse effects reported	25 mg, supports the OSL selected
Healthy adults n = 24	15 mg/day lycopene (as tomato oleoresin or synthetic lycopene, respectively); randomized, controlled	28 days	Significant 2.5 and 2.8-fold increase in serum lycopene, respectively (to 950 and 910 nmol/L), respectively; no adverse effects observed	45 mg, supports the OSL selected
Healthy adults n = 9	20 mg/day lycopene (as tomato paste); randomized, controlled, crossover	11 days	Significant 2.4-fold increase in serum lycopene (to 3430 nmol/L); no adverse effects reported	20 mg, supports the OSL selected
Healthy smoking and nonsmoking adults n = 27	15 mg/day lycopene (as tomato paste); randomized, controlled, crossover	2 week (14 days)	Significant 3.3-fold increase in serum lycopene (to 800 nmol/L); no adverse effects reported	15 mg, supports the OSL selected
Healthy men n = 28	15 mg/day lycopene (as tomato oleoresin); randomized, controlled, crossover	4 week (28 days)	Significant 4-fold increase in serum lycopene (to 750 nmol/L); no adverse effects reported	15 mg, supports the OSL selected
Healthy postmenopausal women n = 8	12 mg/day lycopene (as synthetic lycopene); randomized, controlled	8 week (57 days)	Significant 2.6-fold increase in serum lycopene (to 1300 nmol/L); significant decrease in endogenous DNA damage vs. placebo;	12 mg, supports the OSL selected

			no adverse effects reported	
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*Adapted from Shao and Hathcock (2006)

In GRN 173, several clinical studies of lycopene from different sources have been evaluated. In 12 clinical trials, ranging from 1 to 16 weeks, effects of lycopene (natural and synthetic) have been investigated. In these studies, anthropometric and biochemical measurements, including body weight, full blood counts, and immune function and liver function tests, did not reveal any abnormalities in subjects supplemented with lycopene capsules or lycopene rich foods at levels ranging from 0.5 mg/day for 4 weeks to 75 mg/day for one week. The tolerance of lycopene supplementation was investigated in healthy individuals as well as in prostate cancer patients. Administration of lycopene at doses ranging from 12 to 75 mg/day (approximately 3 to 20 times normal dietary intake) were generally well tolerated.

6.2.5.1. Adverse Effects in Humans

There is not much information on the adverse effects of lycopene in humans. High intakes of lycopene-containing foods has been reported to result in lycopenemia, characterized by an orange discoloration of the skin. All cases of lycopenaemia reported in the literature followed excessive ingestion of lycopene in foods. In a case report, a 61 year old woman who had consumed about 2 L of tomato juice daily for several years reported the incidence of lycopenemia (Reich et al., 1960). Although there was evidence of lycopene and fatty deposits in the liver, there was an absence of measurable hepatic dysfunction. After 3 weeks of consuming a diet free of tomato juice, the orange discoloration faded. Shaw and Koti (2009) reported another case of a 40 year old woman presented with orange discoloration of both palms while the rest of her skin, her sclera and other mucous membranes were normally colored. She consumed tomatoes and tomato-based dishes roughly 5 times per week for 3 years, totaling on average 1.4-1.8 kg of tomatoes weekly. The subject was advised to reduce her tomato intake to 0.5-0.9 kg per week. At a follow-up examination 2 weeks later, the discoloration had faded, and by 4 weeks it had completely disappeared.

Recently, Tanaka et al. (2022) reported a rare case of carotenoderma due to lycopenemia caused by the excessive intake of lycopene-rich vegetables and fruits. These investigators examined a hematoxylin-eosin-stained specimen from the patient and tape-stripped samples by fluorescent microscopy with 488 nm wavelength emission and compared them with normal skin samples. Notably, the patient's samples showed a weaker autofluorescence in the stratum corneum and sweat glands. Furthermore, carotenoid concentrations in the patient's skin was measured noninvasively and a higher score than the average of 24 healthy volunteers was noted. This case further highlights that the excessive intake of lycopene causes carotenoderma and demonstrates that carotenoid deposition is particularly pronounced in the stratum corneum of the skin.

6.2.6. Uses and Regulatory Status

Lycopene has been authorized by the EFSA (E 160d) as a food additive, with an acceptable daily intake of 0.5 mg/kg bw/day, regardless of source (EFSA, 2008). FDA has also approved lycopene use as a natural food and beverage colorant in concentrations within 5 and 70 mg/kg of food (WHO, 2007). In addition, it is included in the Codex General Standard of Food Additives (GSFA) and as such, it can be used in food products under conditions of good manufacturing practices. As per the classification of the WHO for hazard compounds, lycopene belongs to the compounds of “very low” toxicity. Because of the lack of adverse effect data for lycopene in

animals or healthy humans, the Institute of Medicine (IOM, 2000) did not set a tolerable upper intake level for lycopene.

6.2.6.1. FDA Review of Lycopene GRAS Notices

As mentioned earlier, lycopene has been the subject of four GRAS determinations evaluated by FDA and issued no question letters. In the first GRAS notice (GRN 119), BASF Corporation determined that synthetic lycopene is GRAS, through scientific procedures, for use as a direct food ingredient in breakfast cereals, drinks, instant soup, low fat dressings, meal replacements, meatless meat products, nutrient bars, salty snacks, crackers and yogurt at levels ranging from 5 to 70 mg/kg of food. BASF described synthetic lycopene as a crystalline material derived from chemical synthesis and describes a three-stage process for this chemical synthesis. Synthetic lycopene is stored under nitrogen or suspended in an aqueous solution containing antioxidants to prevent oxidation and isomerization of lycopene. The commercial product is prepared as Lycopene 10 Percent (powder spherical particles in a food starch coated matrix of gelatin and sucrose), Lycopene 10 Cold Water Dispersion (CWD), and Lycopene Dispersion 20 Percent (lycopene in vegetable oil). Product specifications include 96% lycopene, limits on total carotenoids, arsenic, lead, copper, zinc, and heavy metals, loss on drying and residue on ignition. BASF estimates the 90th percentile intake of 11 mg/person/day from the proposed uses and is comparable to the intake of lycopene from plant sources.

In the GRAS notice, BASF discusses published and unpublished studies related to the potential toxicity of synthetic and natural lycopene. Based on findings from a 13 week published study a published 13-week oral toxicity study in rats, the NOAEL for synthetic lycopene was determined as 324 mg/kg bw/day. The NOAEL is approximately 4000-fold higher on a body weight basis than the mean EDI of synthetic lycopene. Additionally, no adverse effects were reported in a developmental toxicity study conducted in rats and rabbits fed lycopene products. No mutagenic effects were observed in unpublished genotoxicity studies conducted with BASF's lycopene products. Furthermore, no adverse effects were observed in several published clinical studies conducted with either BASF's lycopene products or natural lycopene. BASF discussed published human studies related to the bioavailability of natural lycopene and synthetic lycopene products. Findings from these studies showed no adverse effects on the absorption of other carotenoids. BASF noted that excessive intake of food or dietary supplements high in carotenoids or lycopene has been associated with carotenodermia and lycopodermia, respectively. Based on the information provided by BASF, as well as other information available to FDA, the agency did not question BASF's conclusion that synthetic lycopene is GRAS under the intended conditions of use.

In addition to the above mentioned first GRAS notice on synthetic lycopene, subsequently FDA received three GRAS notices on lycopene derived from other sources. These included tomato lycopene extract (GRN 156); lycopene from *Blakeslea trispora* (GRN 173); and concentrated tomato lycopene extract (GRN 185). In these GRAS notices, product related information such as description, specification, manufacturing, intended uses and resulting intake, and safety related information on lycopene has been extensively summarized to support the GRAS determination for use of lycopene in food. All these GRAS notices received no question letter from FDA.

Concentrated tomato lycopene extract containing at least 90% lycopene was the subject of GRN 185. In this GRAS notice the notifier discussed safety studies conducted with lycopene, the principal carotenoid of the extract. Additionally, in GRN 185 published and unpublished acute and

subchronic animal toxicity studies, reproductive, developmental and genotoxicity studies, as well as generally available information about the absorption, distribution, metabolism, and elimination of dietary lycopene has been extensively described. Also, published human clinical studies conducted with tomato lycopene extracts have been discussed. The notifier concluded that these studies indicate that long-term consumption of tomato lycopene extract is well-tolerated. Additional safety studies conducted with both synthetic lycopene and lycopene derived from *Blakeslea trispora* are described. In addition to this GRAS notice (GRN 185) and the above mentioned GRN 156, the available safety related information on lycopene derived from *Blakeslea trispora* (subject of GRN 173) and concentrated tomato lycopene extract (GRN 185) discussed the safety related information on lycopene. All these studies described in these GRAS notices, as well as studies appeared subsequent to these GRAS notices, do not raise safety concerns related to the proposed uses of lycopene derived from *S. cerevisiae*, the subject of present GRAS.

6.2.7. Safety of Production Strain

S. cerevisiae has a safe history of use in food and belongs to a species that is known not to produce toxins. *S. cerevisiae* is considered GRAS for use in brewing, baking and winemaking industries (21 CFR §172.896). The genome of *S. cerevisiae* has been extensively studied and it has been determined that the yeast is free of known pathogenicity traits. In addition, several genetically modified *S. cerevisiae* strains for use in fermented beverages have received letters of no objection from the US FDA (GRN 120, GRN 175, GRN 350, GRN 798 and GRN 841). The FDA has not questioned the manufacturer's GRAS conclusion for the intended uses of the following enzyme preparations produced by *S. cerevisiae* strains.

S. cerevisiae species has been included in the Environment Protection Agency (EPA) list as a recipient microorganism that is eligible for exemptions from EPA review under the Toxic Substances Control Act and/or expedited EPA review (40 CFR 725.420). This inclusion is based upon the demonstrated lack of adverse effects for this species. It has been determined that the introduction of genetic material in this species would not increase the potential for adverse effects, provided that the genetic material is limited in size, well-characterized, free of certain sequences and is poorly mobilizable. As per EFSA, yeasts used in food production, particularly bakers/brewer's yeast, are considered among the safest of microorganisms. In Europe, *S. cerevisiae* has been designated Qualified Presumption as Safe (QPS) status, indicating that no additional safety assessment is needed according to the established guidelines. The JZL03 host strain belongs to the *S. cerevisiae* species which has been used for over 7,000 years by humans in fermented beverages. Given this background, it can be concluded that species will remain safe, even after genetic transformation, as long as the transformation does not introduce new or changed capabilities for harmful effects.

As discussed earlier, the subject of this GRAS lycopene is derived from recombinant *S. cerevisiae* strain JZL03. This strain is constructed from *S. cerevisiae* HESHEHG001 and incorporates 13 genes, including 8 target genes and 5 marker genes. As explained in Part 2.5, all transferred gene DNA fragments from recombinant *S. cerevisiae* JZL03 in three batches of the final product lycopene were below the limit of detection. Additionally, the residual protein levels in three batches of the final product, was found to be below the quantitation limit of 17 mg/kg, suggesting the absence of residual protein in the final product lycopene. In addition, Hesheng also assessed the bioinformatics and potential allergenicity of all exogenous proteins:

The similarity in the amino acid sequences of all exogenous proteins expressed by recombinant *S. cerevisiae* strain JZL03 in the and known toxic proteins and anti-nutritional factors in the database is determined by bioinformatics tools. NCBI Entrez database, UniProt database, and BLAST software were used to analysis the toxic proteins and/or anti-nutritional factors. A smaller E value indicates a higher similarity. If search results with an E value ≤ 0.01 was obtained, it is necessary to analyze the specific details of these matches and classify the proteins to determine whether they are toxic proteins or anti-nutritional factors. If the protein under investigation shows high sequence similarity to known toxic proteins or antinutritional factors, further analysis should be conducted on the test protein and the similar sequences, including conserved sequence homology and structural comparisons. According to the sequence alignment results, all exogenous proteins share low sequence homology with known toxic proteins or anti-nutritional factors.

All exogenous protein amino acid sequences are compared for similarity with known allergenic proteins in The Allergen Online Database. A protein is considered to have a higher potential for allergenicity if either of the following conditions is met: (1) the full-length alignment of the exogenous protein has an E-value of less than or equal to 0.01, or if an 80 amino acid segment of the exogenous protein shows 35% or greater sequence homology with a known allergen, or (2) the exogenous protein shares a sequence of 8 consecutive identical amino acids with a known allergen. Otherwise, if none of these conditions are met, the potential for allergenicity is considered low. Based on the comparison results, no high sequence homology was found in all exogenous protein sequences (all results are less than 35%), and no sequence of 8 consecutive amino acids was identified. Therefore, all exogenous proteins introduced by recombinant *S. cerevisiae* strain JZL03 do not show high sequence homology with known allergens, and their potential for allergenicity is considered low.

In summary, the above described information suggest that the production strain used in the manufacturing of lycopene is safe.

6.3. GRAS Panel Evaluation, Summary and Discussion

At the request of Hesheng Technology, an independent panel of recognized experts (At the request of Hesheng Technology, an independent panel of recognized experts (hereinafter referred to as the Expert Panel)¹, qualified by their scientific training and relevant national and international experience to evaluate the safety of food and food ingredients, was convened to assess the Generally Recognized As Safe (GRAS) status of lycopene produced by genetically engineered *Saccharomyces cerevisiae* strain JZL03, when used as a food ingredient and as a nutrient supplement in conventional foods such as baked goods and baking mixes, beverages and beverage bases, breakfast cereals, cheeses, condiments and relishes, confections and frostings, fats and oils, frozen dairy desserts and mixes, gelatins, puddings, and fillings, gravies and sauces, hard candy, milk products, plant protein products, processed fruits and fruit juices, snack foods, soft candy at levels up to 50 ppm and in soups and soup mixes that do not contain tomatoes at levels up to 575 ppm. A comprehensive search of the scientific literature for safety and toxicity information on lycopene and the production strain was conducted through December 2024 and made available to the Expert Panel. The Expert Panel independently and critically evaluated materials submitted by Hesheng Technology and other information deemed appropriate or necessary. Following an

¹Modeled after that described in section 201(s) of the Federal Food, Drug, and Cosmetic Act, As Amended. See also attachments (curriculum vitae) documenting the expertise of the Panel members.

independent, critical evaluation, the Expert Panel conferred on January 22, 2025 and unanimously agreed to the decision described herein.

Hesheng Technology ensured that all reasonable efforts were made to identify and select a balanced Expert Panel with expertise in food safety, toxicology, microbiology, and nutrition. The Expert Panel was selected and convened in accordance with the Food and Drug Administration (FDA)'s guidance for industry on 'Best Practices for Convening a GRAS Panel'². Efforts were placed on identifying conflicts of interest or relevant "appearance issues" that could potentially bias the outcome of the deliberations of the Expert Panel and no such conflicts of interest or 'appearance issues' were identified. The Expert Panel members received a reasonable honorarium as compensation for their time; the honoraria provided to the Expert Panel members were not contingent upon the outcome of their deliberations.

The subject of this GRAS assessment, lycopene preparation is a standardized red to purple red color powder with balsam oriental odor. Lycopene preparation primarily contains lycopene ($\geq 96\%$) that is identical to lycopene found in tomatoes. Lycopene is a type of carotenoid, a natural pigment found in many fruits and vegetables, and is primarily responsible for the red color seen in tomatoes, watermelon, and pink grapefruit. The identity of lycopene has been fully established. The food grade specifications of lycopene preparation, subject of present GRAS, has been established. The purity of lycopene is at least 96%. Specification limits for total lycopene, all trans-lycopene, heavy metals, residual solvents and microbiological contaminants have been established. Hesheng Technology's lycopene preparation is produced as per current Good Manufacturing Processes (cGMP) by recombinant *Saccharomyces cerevisiae* JZL03 constructed from *S. cerevisiae* HESHEHG001 strain. It is manufactured by fermentation. Analytical data from three nonconsecutive lots of lycopene preparation demonstrates that the manufacturing process produces a consistent product that meets the defined specifications.

All raw materials, processing aids, and food contact substances employed in the manufacturing of lycopene preparation are GRAS and/or conform to the specifications stated in 21 CFR and/or the FCC. The genetically engineered *S. cerevisiae* JZL03 strain used by Hesheng Technology in the production of lycopene preparation, is non-toxicogenic, not capable of DNA transfer to other organisms and is a well-characterized fungus with a history of safe use in the food industry. The production organism is constructed through a series of transformations with different expression constructs to enable the production of lycopene. The production organism does not contain any antibiotic resistance genes or plasmids and is not capable of transferring plasmids or antibiotic resistance to other organisms. The use of *S. cerevisiae* JZL03 as the production strain is not expected to result in any safety concerns. Process controls and product specifications are in place to control the levels of residual impurities, as well as heavy metals, microbes, and production organism-derived DNA, ensuring a consistent, food-grade finished ingredient.

Hesheng Technology intends to use lycopene preparation as a food ingredient and as a source of the nutrient in conventional foods such as baked goods and baking mixes, beverages and beverage bases, breakfast cereals, cheeses, condiments and relishes, confections and frostings, fats and oils, frozen dairy desserts and mixes, gelatins, puddings, and fillings, gravies and sauces, hard candy, milk products, plant protein products, processed fruits and fruit juices, snack foods, soft candy at levels up to 50 ppm and in soups and soup mixes that do not contain tomatoes at levels

²Available at:

<https://www.fda.gov/Food/GuidanceRegulation/GuidanceDocumentsRegulatoryInformation/ucm583856.htm>

up to 575 ppm. The intended uses and use levels are same as in GRN 173 for lycopene derived from *Blakeslea trispora*. The proposed use will result in an estimated total daily intake of lycopene from the consumption of foods containing lycopene from *S. cerevisiae* as 5.931 mg/person/day at the mean and as 13.112 mg/person/day at the 90th percentile level. As lycopene is allowed for uses in food, the proposed uses will be substitutional and are unlikely to affect the current existing use or the cumulative intake.

The safety determination of lycopene is based on the totality of the available evidence, including *in vitro* studies, animal experimental studies and human studies of naturally occurring and synthetic lycopene, as well historical and current uses of lycopene as a food ingredient. As lycopene preparation ($\geq 96\%$ pure) produced by *S. cerevisiae*, the subject of this GRAS, is substantially similar to lycopene naturally present in food, derived from tomato and synthetic, the available safety database that exists for lycopene and its preparation is applicable to establish the safety of lycopene preparation. In human diet, the primary source of lycopene is tomatoes, it is also found in fruits like pink grapefruit, water melon, rosehip, papaya and guava. As per 21 CFR 73.585, tomato lycopene extract; tomato lycopene concentrate are approved as color additive. Tomato lycopene is included in EAFUS list. Lycopene from tomatoes (GRN 156; GRN 185), synthetic lycopene (GRN 119) and lycopene synthesized by the fungus *B. trispora* (GRN 173) are permitted for uses in food as GRAS. FEMA has also listed lycopene as GRAS (FEMA No.: 4110).

Given lycopene's chemical (isomeric forms) similarity between the subject of present GRAS and other GRAS notices that received FDA no question letters, the available safety related data from other GRAS notices, as well as other subsequently published safety related studies, are applicable to the subject of this GRAS, lycopene preparation derived from *S. cerevisiae*. In the FDA GRAS notices, published and unpublished acute and subchronic animal toxicity studies, reproductive, developmental and genotoxicity studies, and generally available information about the absorption, distribution, metabolism, and elimination of dietary lycopene have been extensively reviewed and described. In a series of specifically designed unpublished toxicity studies, effects of lycopene preparation (subject of present GRAS) were investigated in acute toxicity, bacterial reverse mutation assay, *in vitro* mammalian chromosome aberration test and *in vivo* mammalian micronucleus test were investigated. All these studies were conducted as per China National Food Safety Standards for such studies. In the acute toxicity study in rats, the LD50 of lycopene preparation was >10 g/kg bw, suggesting that it is practically non-toxic. In the mutagenicity and genotoxicity studies, lycopene preparation did not reveal any potential mutagenic or genotoxic effects.

In multiple published studies that included acute and subchronic animal toxicity, reproductive, developmental and genotoxicity studies, potential adverse effects of lycopene were investigated. In these studies, no adverse effects of test substance containing lycopene were noted. As the study substance containing its active constituent lycopene used for the toxicity studies is chemically similar to the subject of present GRAS, these published studies are applicable to the lycopene preparation GRAS determination and support the safety of the proposed uses of lycopene. In the available literature, there have been some published independent safety assessments on various lycopene products, and four independent 90-day rat studies have been performed. In none of these studies was the NOAEL lower than the highest administered dose. As no adverse effects were observed in any study, use of the highest NOAEL is justified for estimating safe dose.

In multiple human clinical studies with lycopene supplementation (with or without food, based as tomato juice/paste/raw product, or combined with olive oil), various endpoints were

investigated. No adverse effects of lycopene as evaluated by any abnormalities in body weights, hematology, liver function tests or immune function tests in subjects supplemented with lycopene at levels ranging from 0.5 mg/day for 4 weeks, 15 or 30 mg for 3 weeks, 47.1 mg/day for 8 weeks to 75 or 150 mg/day for 1 week. In other studies. The duration of effect studied was considerable and lasted from 30 min to 6 months. In one study, duration was 20 weeks and the lycopene dose was 13.3 mg/day. These studies have not been designed to evaluate the safety of lycopene, but doses were generally well tolerated without reports of any illness or adverse biological effects. None of the clinical trials found an adverse effect related to lycopene administration.

The GRAS assessment for the use of lycopene preparation as a food ingredient and as a nutrient in selected foods is based on the totality of available evidence. All pivotal data and information described in this dossier and used to establish the safety of lycopene preparation under its intended conditions of use are “generally available” (i.e., in the public domain). From the data and information presented herein, Hesheng Technology concludes that lycopene preparation, produced with a genetically engineered strains of *Saccharomyces cerevisiae*, is GRAS for its intended uses in foods, based on scientific procedures.

In summary, there is sufficient qualitative and quantitative scientific evidence, including animal experimental studies, *in vitro* studies, human studies of naturally occurring and synthetic lycopene, as well uses of lycopene as a food ingredient and as a nutrient in selected foods, to assess the safety-in-use of lycopene preparation, produced by recombinant *Saccharomyces cerevisiae* JZL03 strain, the subject of this present GRAS assessment. On the basis of scientific procedures³, the consumption of lycopene preparation produced by fermentation as an added food ingredient in conventional foods such as baked goods and baking mixes, beverages and beverage bases, breakfast cereals, cheeses, condiments and relishes, confections and frostings, fats and oils, frozen dairy desserts and mixes, gelatins, puddings, and fillings, gravies and sauces, hard candy, milk products, plant protein products, processed fruits and fruit juices, snack foods, soft candy at levels up to 50 ppm and in soups and soup mixes that do not contain tomatoes at levels up to 575 ppm, is considered safe. The intended uses are compatible with current regulations, i.e., lycopene preparation is used in foods and beverages, and is produced according to current good manufacturing practices (cGMP).

³ 21 CFR §170.3 Definitions. (h) Scientific procedures include those human, animal, analytical, and other scientific studies, whether published or unpublished, appropriate to establish the safety of a substance.

6.4. Expert Panel Conclusion

Based on a critical evaluation of the publicly available data, summarized herein, the Expert Panel members whose signatures appear below, have individually and collectively concluded that lycopene preparation, produced by recombinant *Saccharomyces cerevisiae* JZL03 strain, meeting the specifications cited herein, and when used as a food ingredient and as a nutrient supplement in conventional foods, such as baked goods and baking mixes, beverages and beverage bases, breakfast cereals, cheeses, condiments and relishes, confections and frostings, fats and oils, frozen dairy desserts and mixes, gelatins, puddings, and fillings, gravies and sauces, hard candy, milk products, plant protein products, processed fruits and fruit juices, snack foods, soft candy at levels up to 50 ppm and in soups and soup mixes that do not contain tomatoes at levels up to 575 ppm, resulting in maximum daily intake of 13.11 mg, is safe.

It is also our opinion that other qualified and competent scientists reviewing the same publicly available toxicological and safety information would reach the same conclusion. Therefore, we have also concluded that lycopene preparation, when used as described, is Generally Recognized As Safe (GRAS) based on scientific procedures.

Signatures



Robert L. Martin, Ph.D.

January 22, 2025
Date



David Ribet, Ph.D.

January 22, 2025
Date



Madhusudan G. Soni, Ph.D., F.A.C.N., F.A.T.S.
Advisor to Expert Panel

January 22, 2025
Date

7. PART VII- LIST OF SUPPORTING DATA AND INFORMATION

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DEPARTMENT OF HEALTH AND HUMAN SERVICES Food and Drug Administration GENERALLY RECOGNIZED AS SAFE (GRAS) NOTICE (Subpart E of Part 170)	Form Approved: OMB No. 0910-0342; Expiration Date: 08/31/2025 (See last page for OMB Statement)	
	FDA USE ONLY	
	GRN NUMBER 001253	DATE OF RECEIPT Feb 22, 2025
	ESTIMATED DAILY INTAKE	INTENDED USE FOR INTERNET
	NAME FOR INTERNET	
KEYWORDS		

Transmit completed form and attachments electronically via the Electronic Submission Gateway (*see Instructions*); OR Transmit completed form and attachments in paper format or on physical media to: Office of Food Additive Safety (*HFS-200*), Center for Food Safety and Applied Nutrition, Food and Drug Administration, 5001 Campus Drive, College Park, MD 20740-3835.

SECTION A – INTRODUCTORY INFORMATION ABOUT THE SUBMISSION

1. Type of Submission (<i>Check one</i>)	
☒ New	☐ Amendment to GRN No. _____ ☐ Supplement to GRN No. _____
2. ☒ All electronic files included in this submission have been checked and found to be virus free. (<i>Check box to verify</i>)	
3. Most recent presubmission meeting (<i>if any</i>) with FDA on the subject substance (<i>yyyy/mm/dd</i>): _____	
4. For Amendments or Supplements: Is your amendment or supplement submitted in response to a communication from FDA? (<i>Check one</i>)	
☐ Yes	If yes, enter the date of communication (<i>yyyy/mm/dd</i>): _____
☐ No	

SECTION B – INFORMATION ABOUT THE NOTIFIER

1a. Notifier	Name of Contact Person Yao Zhi		Position or Title R&D project leader	
	Organization (<i>if applicable</i>) Wuhan Hesheng Technology Co., Ltd.			
	Mailing Address (<i>number and street</i>) Room D102, Building B5, Building B4-B8, Wuhan National Biological Industry (Jiufeng Innovation) Base, No. 666 Gaoxin			
City Wuhan		State or Province Hubei Province		Zip Code/Postal Code
				Country China
Telephone Number +86 27 68789320		Fax Number		E-Mail Address zhiyao@heshengtech.com.cn
1b. Agent or Attorney (if applicable)	Name of Contact Person Wing Yu		Position or Title Technical manager of food division, CIRS GROUP	
	Organization (<i>if applicable</i>) CIRS GROUP USA INC			
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City Arlington		State or Province Virginia		Zip Code/Postal Code 22203
				Country United States of America
Telephone Number +1 703 520 1420		Fax Number		E-Mail Address wing.yu@cirs-group.com

SECTION C – GENERAL ADMINISTRATIVE INFORMATION

1. Name of notified substance, using an appropriately descriptive term

Lycopene

2. Submission Format: *(Check appropriate box(es))*

☒ Electronic Submission Gateway

☐ Electronic files on physical media

☐ Paper

If applicable give number and type of physical media

3. For paper submissions only:

Number of volumes _____

Total number of pages _____

4. Does this submission incorporate any information in CFSAN's files? *(Check one)*

☐ Yes *(Proceed to Item 5)*

☒ No *(Proceed to Item 6)*

5. The submission incorporates information from a previous submission to FDA as indicated below *(Check all that apply)*

☐ a) GRAS Notice No. GRN _____

☐ b) GRAS Affirmation Petition No. GRP _____

☐ c) Food Additive Petition No. FAP _____

☐ d) Food Master File No. FMF _____

☐ e) Other or Additional *(describe or enter information as above)* _____

6. Statutory basis for conclusions of GRAS status *(Check one)*

☒ Scientific procedures *(21 CFR 170.30(a) and (b))*

☐ Experience based on common use in food *(21 CFR 170.30(a) and (c))*

7. Does the submission (including information that you are incorporating) contain information that you view as trade secret or as confidential commercial or financial information? (see 21 CFR 170.225(c)(8))

☐ Yes *(Proceed to Item 8)*

☒ No *(Proceed to Section D)*

8. Have you designated information in your submission that you view as trade secret or as confidential commercial or financial information *(Check all that apply)*

☐ Yes, information is designated at the place where it occurs in the submission

☐ No

9. Have you attached a redacted copy of some or all of the submission? *(Check one)*

☐ Yes, a redacted copy of the complete submission

☐ Yes, a redacted copy of part(s) of the submission

☐ No

SECTION D – INTENDED USE

1. Describe the intended conditions of use of the notified substance, including the foods in which the substance will be used, the levels of use in such foods, and the purposes for which the substance will be used, including, when appropriate, a description of a subpopulation expected to consume the notified substance.

intended for use as a food ingredient and as a nutrient supplement in conventional foods such as baked goods and baking mixes, beverages and beverage bases, breakfast cereals, cheeses, condiments and relishes, confections and frostings, fats and oils, frozen dairy desserts and mixes, gelatins, puddings, and fillings, gravies and sauces, hard candy, milk products, plant protein products, processed fruits and fruit juices, snack foods, soft candy at levels up to 50 ppm and in soups and soup mixes that do not contain tomatoes at levels up to 575 ppm.

2. Does the intended use of the notified substance include any use in product(s) subject to regulation by the Food Safety and Inspection Service (FSIS) of the U.S. Department of Agriculture?

(Check one)

☐ Yes

☒ No

3. If your submission contains trade secrets, do you authorize FDA to provide this information to the Food Safety and Inspection Service of the U.S. Department of Agriculture?

(Check one)

☐ Yes

☐ No, you ask us to exclude trade secrets from the information FDA will send to FSIS.

SECTION E – PARTS 2 -7 OF YOUR GRAS NOTICE

(check list to help ensure your submission is complete – PART 1 is addressed in other sections of this form)

- ☒ PART 2 of a GRAS notice: Identity, method of manufacture, specifications, and physical or technical effect (170.230).
- ☒ PART 3 of a GRAS notice: Dietary exposure (170.235).
- ☒ PART 4 of a GRAS notice: Self-limiting levels of use (170.240).
- ☒ PART 5 of a GRAS notice: Experience based on common use in foods before 1958 (170.245).
- ☒ PART 6 of a GRAS notice: Narrative (170.250).
- ☒ PART 7 of a GRAS notice: List of supporting data and information in your GRAS notice (170.255)

Other Information

Did you include any other information that you want FDA to consider in evaluating your GRAS notice?

☐ Yes ☒ No

Did you include this other information in the list of attachments?

☐ Yes ☐ No

SECTION F – SIGNATURE AND CERTIFICATION STATEMENTS

1. The undersigned is informing FDA that Wuhan Hesheng Technology Co., Ltd.

(name of notifier)

has concluded that the intended use(s) of Lycopene

(name of notified substance)

described on this form, as discussed in the attached notice, is (are) not subject to the premarket approval requirements of the Federal Food, Drug, and Cosmetic Act based on your conclusion that the substance is generally recognized as safe recognized as safe under the conditions of its intended use in accordance with § 170.30.

2. Wuhan Hesheng Technology Co., Ltd. agrees to make the data and information that are the basis for the
(name of notifier) conclusion of GRAS status available to FDA if FDA asks to see them;
agrees to allow FDA to review and copy these data and information during customary business hours at the following location if FDA asks to do so; agrees to send these data and information to FDA if FDA asks to do so.

Room D102, Building B5, Building B4-B8, Wuhan National Biological Industry (Jiufeng Innovation) Base, No. 666 Gaoxin Aven

(address of notifier or other location)

The notifying party certifies that this GRAS notice is a complete, representative, and balanced submission that includes unfavorable, as well as favorable information, pertinent to the evaluation of the safety and GRAS status of the use of the substance. The notifying party certifies that the information provided herein is accurate and complete to the best of his/her knowledge. Any knowing and willful misinterpretation is subject to criminal penalty pursuant to 18 U.S.C. 1001.

3. Signature of Responsible Official,
Agent, or Attorney

Wing Yu

Wing Yu
2025.02.22 21:22:26 +08'00'

Printed Name and Title

Wing Yu, Technical manager of food division, CIRS GROUP

Date (mm/dd/yyyy)

02/22/2025

SECTION G – LIST OF ATTACHMENTS

List your attached files or documents containing your submission, forms, amendments or supplements, and other pertinent information. Clearly identify the attachment with appropriate descriptive file names (or titles for paper documents), preferably as suggested in the guidance associated with this form. Number your attachments consecutively. When submitting paper documents, enter the inclusive page numbers of each portion of the document below.

Attachment Number	Attachment Name	Folder Location (select from menu) (Page Number(s) for paper Copy Only)
	Form3667.pdf	Administrative
	GRASdossier_WuhanHesheng_Lycopene_final.pdf	Administrative

OMB Statement: Public reporting burden for this collection of information is estimated to average 170 hours per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the collection of information. Send comments regarding this burden estimate or any other aspect of this collection of information, including suggestions for reducing this burden to: Department of Health and Human Services, Food and Drug Administration, Office of Chief Information Officer, PRASStaff@fda.hhs.gov. (Please do NOT return the form to this address.). An agency may not conduct or sponsor, and a person is not required to respond to, a collection of information unless it displays a currently valid OMB control number.